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MISO DPP 2019 Cycle 1 East (ATC) Area Study Phase 2 Report

Prepared for MISO

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1.0 EXECUTIVE SUMMARY

Sixteen (16) resource projects have requested to interconnect to the MISO transmission network in the East (ATC) Area and are included in the Definitive Planning Phase 2019 Cycle 1 Phase 2 study (DPP 2019 Cycle 1 Phase 2). All Generating Facilities, except one, have requested both Energy Resource Interconnection Service (ERIS) and Network Resource Interconnection Service (NRIS). J1483 has elected only ERIS.

This report presents the study results of a System Impact Study (SIS) performed to evaluate the generator interconnection requests in the DPP 2019 Cycle 1 Phase 2 study. The study was performed under the direction of Midcontinent Independent System Operator (MISO) by ATC and an ad hoc study group. The results for 2024 scenario are summarized below.

1.1 Project List

The DPP 2019 Cycle 1 Phase 2 has sixteen (16) interconnection requests with a combined nameplate rating of 1,693.06 MW. The DPP 2019 Cycle 1 Phase 2 generator interconnection requests are listed in Table 1.1-1. The one-line diagrams of the interconnection facilities are shown in Appendix C.

Table 1.1-1 – Generating Facilities in DPP April 2019 Phase 2 East (ATC) Area

MISO Queue #	Service Type	Control Area	County, State	Point of Interconnection	Fuel Type	Inverter Model	Main Step-Up Transformer ²	Requested MW	Dispatch (MW) ¹		
									Batteries Discharging		Batteries Charging
									24SH	24SUM	24SH
J1214	NRIS	ALTE	Dane, WI	Rockdale 345 kV	Solar	80 x 4.2 MVA TMEIC PVU-L0840GR Total Rated MVA = 336	Two 345/34.5/13.8 kV 105/140/175 MVA Z=9%	300	0	300	0
J1244	NRIS	UPPCO	Houghton, MI	Winona – Atlantic 69 kV	Wind	8 x 5.3 MVA Siemens Gamesa 5.0-145 (Type 3) Total Rated MVA = 42.4	Two 69/34.5/13.8 kV 15/20/25 MVA Z=7.5%	38.4	38.4	5.9904	38.4
J1251	NRIS	MIUP	Marquette, MI	Perkins – Arnold 138 kV	Solar	28 x 4.2 MVA TMEIC PVU-L0840GR Total Rated MVA = 117.6	One 138/34.5/13.8 kV 69/92/115 MVA Z=9%	100	0	100	0
J1253	NRIS	WPS	Fond du Lac, WI	South Fond du Lac – Fitzgerald 345 kV	Solar	32 x 3.55 MVA Power Electronics HEM FS3430M Total Rated MVA = 113.6	One 345/34.5/13.8 kV 144/240 MVA Z=9%	100	0	100	0
J1304	NRIS	ALTE	Rock, WI	Townline Road – Paddock 138 kV	Solar	22 x 3.3 MVA Power Electronics Freesun FS3000M Total Rated MVA = 72.6	One 138/34.5/13.8 kV 43/72 MVA Z=9%	65	0	65	0
J1305	NRIS	ALTE	Green, WI	Bass Creek – Albany – North Monroe 138 kV	Solar	17 x 3.3 MVA Power Electronics Freesun FS3000M Total Rated MVA = 56.1	One 138/34.5/13.8 kV 33/55 MVA Z=9%	49.9	0	49.9	0
J1316	NRIS	WEC	Kenosha, WI	Paris 138 kV	Battery	14 x 4.2 MVA TMEIC ESS 840 Total Rated MVA = 58.8	One 138/34.5/13.8 kV 36/48/60 MVA Z=9%	50	50	50	-50
J1326	NRIS	ALTE	Dane, WI	Rockdale 345 kV	Battery	20 x 4.2 MVA TMEIC PVU-L0840GR Total Rated MVA = 84	One 345/34.5/13.8 kV 54/72/90 MVA Z=9%	75	75	75	-75
J1345	NRIS	ALTE	Dodge, WI	North Beaver Dam 69 kV	Battery	9 x 3.5 MVA Power Electronics HEC-US FS3000CU15 Total Rated MVA = 34.2	One 69/34.5/13.8 kV 51/85 MVA Z=8.5%	25	25	25	-25
J1370	NRIS	MIUP	Chippewa, MI	Dafer 69 kV	Gas	N/A	One 69/12.47 kV 40/66.8 MVA Z=8%	50	0	50	0
J1374	NRIS	ALTE	Grant, WI	Ebenezer 138 kV	Wind	12 x 6.2 MVA Vestas V162-5.6 (Type 4) Total Rated MVA = 74.4	One 138/34.5/13.8 kV 45/75 MVA Z=10%	67.2	67.2	10.4832	67.2
J1377	NRIS	ALTE	Rock, WI	Rockdale – Wempletown 345 kV	Solar	32 x 3.46 MVA Power Electronics HEM FS3350M Total Rated MVA = 110.72	One 345/34.5/13.2 kV 69/92/115 MVA Z=7%	98.56	0	98.56	0
J1410	NRIS	MGE	Dane, WI	North Madison 345 kV	Solar	80 x 4.2 MVA TMEIC PVU-L0840GR Total Rated MVA = 336	Two 345/34.5/13.8 kV 105/140/175 MVA Z=9%	300	0	300	0

J1411	NRIS	MGE	Dane, WI	North Madison 345 kV	Battery	20 x 4.2 MVA TMEIC PVU-L0840GR Total Rated MVA = 84	One 138/34.5/13.8 kV 54/72/90 MVA Z=9%	75	75	75	-75
J1460	NRIS	ALTE	Rock, WI	Sunrise 138 kV	Solar	60 x 3.59 MVA Power Electronics HEMK 690V FS3450K Total Rated MVA = 215.4	Two 138/34.5/13.8 kV 75/100/125 MVA Z=9%	200	0	200	0
J1483	ERIS	ALTE	Iowa, WI	Hill Valley 345 kV	Wind	44 x 2.653 MVA GE 2.5-127-60 Hz (Type 3) Total Rated MVA = 116.732	One 345/34.5/13.8 kV 72/96/120 MVA Z=9%	99	99	15.444	99

¹ Per MISO BPM 015-r22 the following dispatch assumptions are applied for each Fuel Type.

- a. Combined Cycle (CC) is dispatched to 50% of the Requested MW in the shoulder models, 100% in the summer peak model.
- b. Solar is dispatched offline in the shoulder models, 100% in the summer peak model.
- c. Wind is dispatched to 100% of the Requested MW in the shoulder models, 15.6% in the summer peak model.
- d. Battery is dispatched to 100% of the Requested MW in discharging mode in the summer peak model and dispatched in both charging and discharging modes to +/-100% of the Requested MW in the shoulder models.
- e. Hybrid is dispatched according to Appendix E of MISO BPM 015-r22.

² Three winding transformers add a node not assigned to a PSSE bus number which must be initialized for every model build. Past experience has shown that these nodes not assigned to a PSSE bus number have been a source for power flow model solution difficulties. Since the third winding, when it has no devices such as a shunt or load connected to it, has no impact on the steady-state modeling ATC has opted to not model them. The three-winding transformers will be modeled as a three-winding transformer for the short circuit study.

1.2 Generating Facility Requirements

1.2.1 Voltage Schedule Requirement

ATC requires all generators in its territory to maintain a voltage schedule at the Point of Interconnection (POI). The standard voltage schedule is 1.02 per unit as measured at the POI. This schedule may be changed by the Transmission Operator for specific power plants or specific conditions.

1.2.2 Power Factor Range Requirement

FERC Order 827 and ATC Criteria require all newly interconnecting generators interconnecting to ATC-owned Facilities to provide a power factor range for synchronous and non-synchronous (e.g., wind turbines, solar) generation of 0.95 leading (when a Generating Facility is consuming reactive power from the transmission system) to 0.95 lagging (when a Generating Facility is supplying reactive power to the transmission system).

The Generating Facility must be capable of maintaining ATC's standard power factor range at all power output levels by providing continuous dynamic reactive power at the following locations (i.e. point of measurement):

- A. The POI for all synchronous generators
- B. The high-side of the generator substation for all non-synchronous generators

This power factor requirement applies at all power output levels unless the Generating Facility is physically disconnected from the ATC transmission system. Physically disconnected means an open

- generator step-up transformer high or low-side breaker(s) for a synchronous Generation Facility, or
- generator substation step-up transformer high-side breaker(s) for a nonsynchronous Generation Facility.

For synchronous generators, the interconnection studies will account for the net effect of all energy production devices and losses on the Customer's side of the POI.

- Dynamic reactive power provided by a synchronous Generating Facility may meet the dynamic reactive power requirement by utilizing a combination of the inherent dynamic reactive power capability of the synchronous generator, dynamic reactive power devices (e.g., Static VAR Compensators), and static reactive power devices (e.g., capacitors) to make up for losses.
- Dynamic leading reactive power provided by a synchronous Generating Facility cannot use inductive losses from generator step-up transformer(s) and generator tie line(s) to meet the leading power factor calculation at POI. A synchronous Generating Facility must be able to meet a 0.95 leading power factor, as measured at the generator terminal (i.e. the low side of the generator step-up transformer).

For non-synchronous generators, the interconnection studies will account for the net effect of all energy production devices and losses on the Customer's side of generator substation step-up transformer.

- Dynamic reactive power provided by a non-synchronous Generating Facility must meet the following requirement from FERC order 827 paragraph 35:
 - “Non-synchronous generators may meet the dynamic reactive power requirement by utilizing a combination of the inherent dynamic reactive power capability of the inverter, dynamic reactive power devices (e.g., Static VAR Compensators), and static reactive power devices (e.g., capacitors) to make up for losses.”
- Dynamic leading reactive power provided by a non-synchronous Generating Facility cannot use inductive losses from pad-mount and station step-up transformers and collector system to meet the leading power factor calculation at the high-side of the generator substation. A non-synchronous Generating Facility must be able to meet a leading 0.95 power factor, as measured at the generator terminal (i.e. the low side of the pad-mount transformer).
- Dynamic lagging reactive power provided by a non-synchronous Generating Facility cannot use collector system charging to meet the lagging power factor calculation at the high-side of the generator substation. A non-synchronous Generating Facility must be able to meet a lagging 0.95 power factor, as measured at the generator terminal (i.e. the low side of the pad-mount transformer).
- When the Generating Facility is not generating active power (i.e. zero MW output):
 - a. The reactive power injection to the transmission system at the high-side of the generator substation should be zero Mvar. Exceptions for a short transition period during the switching of inverter control modes may be granted on a case by case basis if there is no reliability concern. For example, solar PV facilities may have a transition period during sunrise and sunset periods. The transition period should be minimized by the non-synchronous Generating Facilities as much as possible.
 - b. When the Generating Facility is physically connected but operating at zero MW and zero Mvar as measured at the high-side of the generator substation, the Generating Facility is not required to control system voltage.
 - c. The requirements in bullet a and b above are minimum requirements. If a nonsynchronous Generating Facility has the capability to provide dynamic voltage control and maintain ATC's voltage schedule when it is at zero MW output, that is acceptable and preferred by ATC.

Static reactive power devices (e.g., capacitors and inductors) can only be used to make up for

- Inductive losses between the generator terminal and the POI for synchronous generators, or
- Inductive losses or collector system charging between the generator terminal and the high side of generator substation for non-synchronous generators.

All other reactive power needed to meet the power factor requirement must be provided by continuous and sustainable dynamic sources. Operation across the entire power factor range must be fully dynamic, variable, and capable of sustained indefinite operation.

Static sources can be switched on or off in the range of seconds and provide reactive power in large discrete blocks. Cap Banks are considered static sources of reactive power.

Dynamic sources can provide variable amounts of reactive power in a few milliseconds. Static Var Compensators (SVCs), Static Synchronous Compensators (STATCOMs), Flexible AC Transmission Systems (FACTS), inverters, and synchronous condensers are all considered dynamic sources of reactive power.

For non-synchronous generation projects in the DPP 2019 Cycle 1 Phase 2 study group, if they did not have a signed Generator Interconnection Agreement (GIA) or Provisional GIA (PGIA) by September 21, 2016, they are required to provide dynamic reactive power within the range of 0.95 leading to 0.95 lagging at the high-side of the generator substation.

The generation requests shown in Table 1.2.2-1 did not meet the static reactive power requirements per ATC Criteria at the time of the model completion and are required to provide additional static reactive power sources. All other requests in this queue met FERC Order 827 and ATC Criteria for static reactive power requirements.

Table 1.2.2-1 – Additional Static Mvar to meet ATC Capacitive Power Factor and FERC Order 827 Power Factor Requirements

MISO Queue #	Type	Additional Static Shunt Compensation ¹ (Mvar)
J1251	non-synchronous	3.2
J1304	non-synchronous	6.2
J1305	non-synchronous	4.7
J1483	non-synchronous	18.2

¹ Additional compensation is required to meet the criteria at the POI Bus for synchronous Generating Facilities or the high-side of the generator substation for non-synchronous Generating Facilities.

When non-synchronous Generating Facility is not generating active power (i.e. zero MW output), the reactive power injection to the transmission system at the high-side of the generator substation should be zero Mvar. Table 1.2.2-2 shows

the Mvar levels expected at the high-side of the generator substations for all non-synchronous generation requests when not generating. Inductive power devices or the generating facilities inverters can be used to counter var flows created at low or no output levels.

Table 1.2.2-2 – Assessment of Reactive Power Injection to the Transmission System When Non-synchronous Generator is at Zero MW Output

MISO Queue #	Additional inductive Mvar required at the high-side of generator substation to achieve zero reactive power injection when non-synchronous generator is at zero MW output
J1214/J1326	35.4/22.9
J1244	0.8
J1251	0.7
J1253	0.1
J1304	0.6
J1305	0.5
J1316	0.3
J1345	4.5
J1370	0.3
J1374	3.1
J1377	2.8
J1410/J1411	45/22.9
J1460	8
J1483	2.6

1.2.3 Island Detection and Operation

In circumstances where the Generating Facility has no governor controls and the transmission system design could result in an islanding condition for the outage of two transmission elements, ATC requires the Customer to implement additional protection systems as mutually agreed by the Customer and ATC to prevent generation from being isolated or islanded with interconnected load. Alternatively, ATC will require the Customer to curtail their generation for circumstances that could result in an island condition with the next contingency.

This would apply to the following Generating Facilities from this DPP cycle that may lack adequate governor controls to safely and reliably sustain an island with load.

- J1305
- J1374

- Until completion of Lancaster transformer replacement & substation rebuild (MTEP PID 20925)

1.3 Total Network Upgrades for all Projects

The cost allocation of Network Upgrades for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of Energy Resource Interconnection Service and Network Resource Interconnection Service as of the SIS report date. The total cost of Network Upgrades required for each generator interconnection request is listed in Table 1.3-1. The costs for Network Upgrades are planning level estimates and subject to be revised in the facility studies. No Network Upgrade projects driven by stability, short circuit and PJM Affected System analysis were not identified since these analyses were not performed in Phase 1. All Interconnection Facility Project Diagrams are documented in Appendix C and all Network Upgrade Project Diagrams are documented in Appendix D (No project diagrams are developed for line upgrades).

Table 1.3-1 – Total Cost of Network Upgrades for DPP 2019 Cycle 1 Phase 2 Generator Interconnection Requests

MISO Queue #	Requested MW	ERIS Network Upgrades (\$)				NRIS Network Upgrades (\$)	Interconnection Facilities (\$)		Shared Network Upgrade (\$)	Total Cost of Network Upgrades (Exclude TOIF & Affected Systems) (\$)	M2 Received (\$) ¹	M3 Received (\$) ²	M4 Due (\$) ⁷
		Steady - State Thermal & Voltage	Transient Stability	Short Circuit	Affected System		TO Network Upgrades	TO-Owned Direct Assigned (TOIF)					
[a]	[b]	[c]	[d]	[e]	[f]	[g]	[h]	[i]	[j]	[k]=[c]+[d]+[e]+[g]+[h]	[l] = \$4000 x [b]	[m]= (10%of [k])-[l]	[n]= (20%of [k])-[l] –[m]
J1214	300	0	0	0	89,420,000	1,964,373	1,359,849	675,120	0	3,324,222	1,200,000	0	0
J1244	38.4	0	0	0	0	964,627	6,841,936	274,300	0	7,806,563	153,600	543,364	864,349
J1251	100	0	0	0	0	0	7,623,741	378,914	0	7,623,741	400,000	0	1,124,748
J1253	100	0	0	0	0	670,754	11,991,921	629,915	0	12,662,675	400,000	1,416,526	716,009
J1304	65	0	0	0	0	4,251,106	8,749,504	363,168	0	13,000,610	260,000	990,187	1,349,935
J1305	50	0	0	0	0	3,173,359	7,766,049	366,366	0	10,939,408	200,000	976,987	1,010,895
J1316	50	0	0	0	0	292,629	0	0	0	292,629	200,000	0	0
J1326	75	0	0	0	0	490,689	1,359,849	675,120	0	1,850,538	300,000	0	70,108
J1345	25	0	0	0	0	19,404,093	0	0	0	19,404,093	100,000	1,909,687	1,871,132
J1370	50	325,098	0	0	0	25,477,000	0	0	0	25,802,098	200,000	2,380,210	2,580,210
J1374	67.2	0	0	0	0	3,598,091	1,201,495	375,518	0	4,799,586	380,000	292,371	287,546
J1377	98.56	0	0	0	38,720,000	549,405	13,937,272	637,088	0	14,486,677	394,240	1,001,558	1,501,538
J1410	300	0	0	0	0	2,273,562	4,436,635	580,990	0	6,710,197	1,200,000	262,431	0
J1411	75	0	0	0	0	567,933	4,436,635	580,990	0	5,004,568	300,000	465,989	234,924
J1460	200	0	0	0	324,000	26,681,318	4,065,157	400,108	0	30,746,475	800,000	2,595,199	2,754,096
J1483	99	0	0	0	0	0	1,613,040	639,848	0	1,613,040	396,000	573,239	0
Total (\$)	-	325,098	0	0	128,464,000	90,358,940	75,383,082	6,577,445	0	166,067,120	6,883,840	13,407,748	14,365,489

¹ M2: Milestone Payment dollars received by MISO² M3 = (10% of NU)-M2³ ATC projects are estimated in ISD dollars with a 2.5% annual escalation rate and include a contingency based on a project risk generally between 5% – 20%.⁴ Pre-Certification costs are not included.⁵ Transmission Owner shall also collect from Interconnection Customer a tax gross-up amount on the payments made to Transmission Owner using the Transmission Owner rate in effect at the time the payment is received from Interconnection Customer. The current Transmission Owner tax gross-up rate is 12.848%.⁶ TOIF: Non-reimbursable Transmission Owner Interconnection Facilities to which tax gross-up amount must also be applied.⁷ M4= (20% of NU)-M2-M3.

1.4 Stability Operating Restrictions Summary

For NERC Category P6 stability constraints, operating restrictions may be required under prior outage conditions to avoid instability caused by the next event. A summary of all P6 stability operating restrictions identified for study generators in the DPP 2019 Cycle 1 Phase 2 are provided in Table 1.4-1. Maximum allowed real power outputs to mitigate P6 stability constraints under prior outage conditions were identified for each generating facility. These constraints are discussed in further detail in Sections 3 and 6.

Table 1.4-1 – NERC Category P6 Operating Restriction Analysis Result Summary

MISO Queue #	Max MW	NERC Category P6 (N-1-1) Event		TSAT Max MW Allowed ¹	PSCAD Max MW Allowed ¹
		1 st Contingency	2 nd Contingency		
J1244/ Mihm	38.4/ 54.9			4/NA	3.8/NA
				22/NA	0/NA
				0/39	19.1/39 (3 units)
					15.3/31 (2 units)
					38.4/18.8 (1 unit)
				NA/39	not run
J1370	50			28	not run
J1374	67.2			0	NA

¹NA stands for no restriction.

1.5 In-Service Dates and Cost Estimates

ATC understands that the estimated in-service date may not align with the Interconnection Customer's Synchronization Date; however, negotiated and executed agreements, such as an Engineering and Procurement Agreement, can be used prior to the GIA execution date to expedite Network Upgrades. In absence of any special arrangement, typical times to develop a new Interconnection Facility is about 24-36 months after the GIA is executed, assuming no delays due to Interconnection Customer's permits, state processes, land acquisitions, deliverables (such as a finish graded substation site, etc.) It also assumes that system outages required to construct facilities can be obtained timely. The cost estimates for Interconnection Facilities are based on the in-service date provided in the Interconnection Customer's application data. Therefore, any change in in-service date will have impact on the cost estimates. The requested dates for Interconnection Facility in-service, synchronization, and commercial operation are summarized in Table 1.5-1.

Table 1.5-1 – Requested Interconnection Facilities In-Service Dates, Synchronization Dates and Commercial Operation Dates

MISO Queue #	Requested Interconnection	Requested Synchronization Date	Requested Commercial Operation Date
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	Facility In-service Date		
J1214	October 30, 2022	October 30, 2022	December 31, 2022
J1244	September 15, 2021	October 5, 2021	October 29, 2021
J1251	October 30, 2022	October 01, 2022	December 31, 2022
J1253	August 1, 2022	September 1, 2022	December 31, 2022
J1304	December 1, 2021	January 2, 2022	May 1, 2022
J1305	December 1, 2021	January 2, 2022	May 1, 2022
J1316	October 30, 2022	October 30, 2022	December 31, 2022
J1326	October 30, 2022	October 30, 2022	December 31, 2022
J1345	October 1, 2021	October 15, 2021	December 1, 2021
J1370	March 1, 2021	April 1, 2021	June 1, 2021
J1374	August 1, 2023	October 1, 2023	December 1, 2023
J1377	July 15, 2022	August 15, 2022	October 31, 2022
J1410	October 30, 2022	October 30, 2022	December 31, 2022
J1411	October 30, 2022	October 30, 2022	December 31, 2022
J1460	June 30, 2022	August 31, 2022	October 31, 2022
J1483	October 30, 2023	October 1, 2023	December 31, 2023

1.6 MTEP Projects

If a MTEP transmission project(s) resolves the constraint, and that project(s) is approved by the Board within (1) calendar year of the GIA execution or execution of an amendment thereof, then the Interconnection Customer will not be responsible for transmission upgrade(s) that would resolve the constraint. If that MTEP project(s) is not approved within one (1) calendar year of the GIA execution or execution of an amendment thereof, the Interconnection Customer will be responsible for those transmission upgrade(s).

1.7 Further Study

The next step in the MISO Generator Interconnection Procedures is to perform additional SISs (if needed), Interconnection Customer Interconnection Facility Studies, and Network Upgrade Facility Studies. Those Facilities Studies will specify in more detail the time and cost of the equipment, engineering, procurement, and construction of the Interconnection Facilities and Network Upgrades identified in this report.

1.8 Compliance Summary

This study report partially meets NERC TPL-001-4 standard, FAC-002-2 standard, and Local Planning Criteria. In ATC's annual Ten-Year Assessment (TYA) and MISO annual MTEP studies, additional compliance related studies will be

performed for the generator interconnection requests with signed GIAs. Appendix J describes in detail the NERC and Local Criteria requirements met by this SIS report.

2.0 STEADY-STATE ANALYSIS

Steady-state analysis was performed to identify thermal and voltage upgrades required to interconnect the generator interconnection requests in the DPP 2019 Cycle 1 Phase 2 to the transmission system. Detailed study assumptions, criteria, and methodology are documented in Appendix A.

2.1 Model Development

2.1.1 Study Cases

Three study cases for the steady-state thermal and voltage analysis were developed based upon the expected topology for the local area: 2024 summer peak batteries discharging, 2024 shoulder batteries discharging, and 2024 shoulder batteries charging. The ATC system in these cases was updated with the most recent data available at the time of model construction. The cases were reviewed by ATC and the Interconnection Customers. Based on this review, the cases were further modified to account for model updates, changes, and competing generation requests that had dropped out of the MISO queue since the models were built.

The prior queued generator interconnection requests in the ATC system that are included in the study cases are listed in Table 2.1.1-1. Associated Network Upgrades were also included based on their expected in-service date.

Table 2.1.1-1 – Prior Queued Generator Interconnection Requests Not Yet In-Service at the Time of Model Development

MISO Queue #	Type	Control Area	Requested MW
J732	CC	WPS	561.5
J818	Solar	WEC	149
J849	Solar	MIUP	125
J850	Solar	ALTE	250
J855	Wind	ALTE	100
J864	Solar	ALTE	49.98
J871	Solar	ALTE	100
J878	Solar	WEC	200
J947	Solar	ALTE	200
J986	Solar	ALTE	149.76
J1000	Solar	ALTE	50
J1002	Solar	ALTE	99
J1003	Solar	ALTE	50
J1042	Solar	ALTE	180
J1101	Battery	WPS	20
J1153	Solar	WEC	150
J1154	Solar	WEC	75
J1171	Solar	WEC	100
J1183	Solar	MIUP	1.35
J1188	Solar	ALTE	50

Public information related to the MISO Generator Interconnection Request queue can be found at:

https://www.misoenergy.org/planning/generator-interconnection/GI_Queue/

The summer peak discharging, shoulder discharging, and shoulder charging cases dispatched generation within MISO according to section 6.1.1.1.2, Study Case Development, in the MISO BPM-015-r22.

All excess generation from this methodology is dispatched against all units in MISO Classic proportionally, excluding the units in the current DPP cycle. Scheduled firm transfers are ignored in this dispatch methodology.

2.1.2 Benchmark Cases

Two benchmark cases were used to benchmark system performance without the DPP 2019 Cycle 1 Phase 2 generating facilities and were created by taking the DPP 2019 Cycle 1 Phase 2 Generating Facilities offline from the corresponding study cases. The MISO Classic was used for power balance, where generation was scaled in proportion to P_{max} minus P_{gen} .

2.2 Reactive Power Requirements (FERC Order 827)

All synchronous and non-synchronous generation in this queue were evaluated to determine if the requests meet FERC Order 827 and ATC Planning Criteria. Refer to PLG-METH-0005 in Appendix B for details on the methodology used to determine power factor compliance. All of the reactive resources modeled in the assessment are summarized in Table 2.2-1.

Table 2.2-1 – Reactive Resources Modeled in Generator Interconnection Power Factor Analysis

MISO Queue #	Machines				Dynamic Devices			Static Devices		
	Description	Real Power (MW)	Capacitive Reactive Power (Mvar)	Inductive Reactive Power (Mvar)	Description	Capacitive Reactive Power (Mvar)	Inductive Reactive Power (Mvar)	Description	Capacitive Reactive Power (Mvar)	Inductive Reactive Power (Mvar)
J1214/J1326	Solar Machines/Battery Storage	375	181.62	-181.62	None	N/A	N/A	2-4 MVAR Cap Bank	8	0
J1244	Wind Turbines	38.4	19.4	-19.4	None	N/A	N/A	None	0	0
J1251	Solar Machine	105.84	51.26	-51.26	None	N/A	N/A	1-3 MVAR Cap Bank	3	0
J1253	Solar Machine	101	52	-52	None	N/A	N/A	None	0	0
J1304	Solar Machine	65	29.8	-29.8	None	N/A	N/A	None	0	0
J1305	Solar Machine	49.9	22.88	-22.88	None	N/A	N/A	None	0	0
J878/J1316	Solar Machines/Battery Storage	252.92	141.09	-141.09	None	N/A	N/A	None	0	0
J1003/J1345	Solar Machines/Battery Storage	77	45.89	-45.89	None	N/A	N/A	None	0	0
J1370	Simple Cycle	50	28	-24.68	None	N/A	N/A	None	0	0
J1374	Wind Turbines	67.2	31.97	-23.84	None	N/A	N/A	2-2 MVAR Cap Bank	4	0
J1377	Solar Machine	98.56	50.494	-50.494	None	N/A	N/A	None	0	0
J1410/J1411	Solar Machines/Battery Storage	375	181.62	-181.62	None	N/A	N/A	2-4 MVAR Cap Bank	8	0
J1460	Solar Machines	200	65.74	-65.74	None	N/A	N/A	2-26 MVAR Cap Bank	52	0
J1483	Wind Turbines	100.88	36.44	-36.44	None	N/A	N/A	1-3 MVAR Cap Bank	3	0

The dynamic capacitive power factor requirement analysis showed all requests meeting ATC Criteria and FERC Order 827 requirements. The results are summarized in Table 2.2-2.

Table 2.2-2 – Assessment of Dynamic Capacitive Power Factor Requirement

MISO Queue #	Machine Terminal Bus #	Capability at Machine Terminal		Additional Dynamic Capacitive Reactive Power ¹ (Mvar)	Dynamic Power Factor Provided	Meets Requirement ² ?	Additional Requirement ³ (Mvar)
		Real Power (MW)	Capacitive Reactive Power (Mvar)				
J1214/J1326	43260, 43261, 43262	375.0	181.62	0.0	0.90	Yes	0.0
J1244	42440, 42441	38.4	19.400	0.0	0.89	Yes	0.0
J1251	42520	105.84	51.26	0.0	0.90	Yes	0.0
J1253	42530	101.0	52.000	0.0	0.89	Yes	0.0
J1304	43040	65.0	29.800	0.0	0.91	Yes	0.0
J1305	43054	49.9	22.880	0.0	0.91	Yes	0.0
J878/J1316	88784, 88784, 43160	252.9	141.000	0.0	0.87	Yes	0.0
J1003/J1345	40030, 43450	77.0	45.900	0.0	0.86	Yes	0.0
J1370	43700	50.0	28.000	0.0	0.87	Yes	0.0
J1374	43740	67.2	31.97	0.0	0.90	Yes	0.0
J1377	43770	98.56	50.494	0.0	0.89	Yes	0.0
J1410/J1411	44110, 44111, 44112	375.0	181.62	0.0	0.90	Yes	0.0
J1460	44600, 44601	200	65.74	0.0	0.95	Yes	0.0
J1483	44830	110.88	36.44	0.0	0.95	Yes	0.0

¹ Dynamic capacitive reactive power provided by Interconnection Customer owned equipment in addition to the machine.

² ATC requires a 0.95 ATC Capacitive Dynamic Power Factor.

³ Additional dynamic reactive power required to meet ATC Capacitive Dynamic Power Factor.

The static capacitive power factor requirement analysis showed 4 of the 17 requests do not meet the ATC Criteria or FERC Order 827 requirements. The results are summarized in Table 2.2-3.

Table 2.2-3 – Assessment of Static Capacitive Power Factor Requirement

MISO Queue #	Point of Measurement	Capability at Point of Measurement ¹		Power Factor	Meets Requirement?	Additional Requirement (Mvar)
		Real Power (MW)	Reactive Power (Mvar)			
J1214/J1326	43269	368.6	158.5	0.92	Yes	0.0
J1244	42446	37.7	12.7	0.95	Yes	0.0
J1251	42526	104.7	31.2	0.95	No	3.2
J1253	42534	99.9	34.7	0.94	Yes	0.0
J1304	43043	64.1	14.9	0.97	No	6.2
J1305	43053	49.1	11.4	0.97	No	4.7
J878/J1316	88780	248.9	89.0	0.94	Yes	0.0
J1003/J1345	40033	75.5	30.7	0.93	Yes	0.0
J1370	43703	48.1	17.2	0.94	Yes	0.0
J1374	43743	65.14	20.49	0.95	Yes	0.0
J1377	43773	96.7	32.25	0.95	Yes	0.0
J1410/J1411	44119	368.8	167.5	0.91	Yes	0.0
J1460	44605	196.85	85.60	0.92	Yes	0.0
J1483	44833	108.64	17.52	0.99	No	18.2

¹ Point of Measurement is the POI Bus for synchronous machines and high side of generator substation for asynchronous machines.

The dynamic inductive power factor requirement analysis showed all requests meet ATC Criteria and FERC Order 827 requirements. The results are summarized in Table 2.2-4.

Table 2.2-4 – Assessment of Dynamic Inductive Power Factor Requirement

MISO Queue #	Machine Terminal Bus #	Capability at Machine Terminal		Additional Dynamic Inductive Reactive Power ¹ (Mvar)	Dynamic Power Factor Provided	Meets Requirement ² ?	Additional Requirement ³ (Mvar)
		Real Power (MW)	Inductive Reactive Power (Mvar)				
J1214/J1326	43260, 43261, 43262	375.0	-181.62	0.0	0.90	Yes	0.0
J1244	42440, 42441	38.4	-19.4	0.0	0.89	Yes	0.0
J1251	42520	105.84	-51.26	0.0	0.90	Yes	0.0
J1253	42530	101.0	-52	0.0	0.89	Yes	0.0
J1304	43700	65.0	-29.8	0.0	0.91	Yes	0.0
J1305	43700	49.9	-22.88	0.0	0.91	Yes	0.0
J878/J1316	88784, 88784, 43160	252.9	-141	0.0	0.87	Yes	0.0
J1003/J1345	40030, 43450	77.0	-45.9	0.0	0.86	Yes	0.0
J1370	43700	50.0	-24.7	0.0	0.90	Yes	0.0
J1374	43740	67.2	-23.84	0.0	0.94	Yes	0.0
J1377	43770	98.56	-50.494	0.0	0.89	Yes	0.0
J1410/J1411	44110, 44111, 44112	375.0	-181.62	0.0	0.90	Yes	0.0
J1460	44600, 44601	200.0	-65.74	0.0	0.95	Yes	0.0
J1483	44830	110.88	-36.44	0.0	0.95	Yes	0.0

¹ Dynamic inductive reactive power provided by Interconnection Customer owned equipment in addition to the machine.

² ATC requires a 0.95 ATC Inductive Dynamic Power Factor.

³ Additional dynamic reactive power required to meet ATC Inductive Dynamic Power Factor.

The static inductive power factor requirement analysis showed all requests meeting ATC Criteria and FERC Order 827 requirements. The results are summarized in Table 2.2-5.

Table 2.2-5 – Assessment of Static Inductive Power Factor Requirement

MISO Queue #	POI Bus (synchronous) or HV Bus (asynchronous)	MW at POI (synchronous) or HV Bus (asynchronous)	Mvar at POI (synchronous) or HV Bus (asynchronous)	POI (synchronous) or HV Bus (asynchronous) Power Factor	Meets 0.95 ATC Inductive Power Factor Requirement at POI Bus (synchronous) or HV Bus (asynchronous)?
J1214/J1326	43269	368	-215.9	0.86	Yes
J1244	42446	37.5	-28	0.80	Yes
J1251	42526	104.5	-78.7	0.80	Yes
J1253	42534	99.8	-70.8	0.82	Yes
J1304	43043	63.9	-48.1	0.80	Yes
J1305	43053	48.9	-37	0.80	Yes
J878/J1316	88780	248.0	-205.7	0.77	Yes
J1003/J1345	40033	74.5	-75.9	0.70	Yes
J1370	43703	47.8	-37.2	0.79	Yes
J1374	43743	64.90	-41.14	0.84	Yes
J1377	43773	96.4	-72.17	0.80	Yes
J1410/J1411	44119	368.5	-204.7	0.87	Yes
J1460	44605	196.87	-96.21	0.90	Yes
J1483	44833	108.5	-59.42	0.88	Yes

Interconnection Customers must comply with the power factor requirements while interconnected, including when not generating or at low output, up to the capability of the generating facility. These requirements apply at all times unless the point of measurement (high side of the main power transformer for asynchronous plants and POI for synchronous plants) is physically disconnected from the ATC transmission system by an open circuit. Inductive power devices or the generating facilities inverters can be used to counter MVARs created at low or no output levels. Table 2.2-6 shows the Mvar levels expected for each generation request when not generating.

Table 2.2-6 – Assessment of Reactive Power Injection to the Transmission System When Non-synchronous Generator is at Zero MW Output

MISO Queue #	Additional Inductive Mvar required at POI Bus (synchronous) or HV Bus (asynchronous) to meet ATC Power Factor Requirement when Generation Request is not Generating
J1214/J1326	35.4/22.9
J1244	0.8
J1251	0.7
J1253	0.1
J1304	0.6
J1305	0.5
J1316	0.3
J1345	4.5
J1370	0.3
J1374	3.1
J1377	2.8
J1410/J1411	45/22.9
J1460	8
J1483	2.6

2.3 NERC TPL Contingency Analysis Results

The incremental impact of the proposed generator interconnection on transmission facilities was evaluated by comparing steady state power flows and voltages between benchmark cases (without DPP 2019 Cycle 1 Phase 2 projects) and study cases (with DPP 2019 Cycle 1 Phase 2 projects). Post-contingency cases were solved with transformer tap adjustment enabled, area interchange adjustment disabled, phase shifter adjustment enabled, and switched shunt adjustment enabled. Detailed NERC TPL Category P contingencies that were studied are described in Table A.2.1-1 in Appendix A.

2.3.1 2024 Summer Peak Batteries Discharging

The study identified the steady-state thermal and voltage constraints that qualified as MISO Injection Constraints in the 2024 Summer Peak Discharging study model under NERC Category P0-P7 Planning Events (except NERC Category P3 and P6). Detailed steady-state power flow results and Injection Constraint determination can be found in Appendix E. A summary of the 2024 Summer Peak Discharging MISO Injection Constraints that require Network Upgrades is presented in Table 2.3.1. No voltage constraints were identified in the steady-state analyses.

Table 2.3.1 – 2024 Summer Peak Batteries Discharging Steady-State Injection Constraints Requiring Network Upgrades

Study Case	Overloaded Facility	Facility Owner	Resulted by "No Load Loss Allowed" ¹ NERC TPL Planning Events	Responsible Generator(s)
2024 Summer Discharging	Dafter – Nine Mile 69 kV line	ATC	Base Case P11 P12 P21	J1370

¹ The "No Load Loss Allowed" NERC TPL Planning Events refer to all the Planning Events in NERC TPL-001-4 Table 1 that the interruption of Firm Transmission Service and Non-Consequential Load Loss are not allowed.

² Per ATC's Planning Criteria, for Planning Events that allow Non-Consequential load loss, short duration exposure to potential post-contingency overload represents an acceptable risk assuming system adjustment within that short period of time can eliminate the exceedance and restore facility loading within Long-Term emergency ratings.

2.3.2 2024 Shoulder Batteries Discharging

The study identified the steady-state thermal and voltage constraints that qualified as MISO Injection Constraints in the 2024 Shoulder Discharging study model under NERC Category P0-P7 Planning Events (except NERC Category P3 and P6). Detailed steady-state power flow results and Injection Constraint determination can be found in Appendix E. A summary of the 2024 Shoulder Discharging MISO Injection Constraints that require Network Upgrades is presented in Table 2.3.2. No voltage constraints were identified in the steady state analyses.

Table 2.3.2 – 2024 Shoulder Batteries Discharging Steady-State Injection Constraints Requiring Network Upgrades

Study Case	Overloaded Facility	Facility Owner	Resulted by "No Load Loss Allowed" ¹ NERC TPL Planning Events	Responsible Generator(s)
2024 Shoulder Discharging	J1244 POI – Portage Tap – Atlantic 69 kV line	ATC	Base Case P12 P13 P14 P21 P23	J1244
	J1244 POI – Winona 69 kV line	ATC	P12 P21	J1244

¹ The "No Load Loss Allowed" NERC TPL Planning Events refer to all the Planning Events in NERC TPL-001-4 Table 1 that the interruption of Firm Transmission Service and Non-Consequential Load Loss are not allowed.

² Per ATC's Planning Criteria, for Planning Events that allow Non-Consequential load loss, short duration exposure to potential post-contingency overload represents an acceptable risk assuming system adjustment within that short period of time can eliminate the exceedance and restore facility loading within Long-Term emergency ratings.

2.3.3 2024 Shoulder Batteries Charging

The study identified the steady-state thermal and voltage constraints that qualified as MISO Injection Constraints in the 2024 Shoulder Charging study model under NERC Category P0-P7 Planning Events (except NERC Category P3 and P6). Detailed steady-state power flow results and Injection Constraint determination can be found in Appendix E. A summary of the 2024 Shoulder Charging MISO Injection Constraints that require Network Upgrades is presented in Table 2.3.2-4. No voltage constraints were identified in the steady state analyses.

Table 2.3.4 – 2024 Shoulder Batteries Charging Steady-State Injection Constraints Requiring Network Upgrades

Study Case	Overloaded Facility	Facility Owner	Resulted by "No Load Loss Allowed" ¹ NERC TPL Planning Events	Responsible Generator(s)
2024 Shoulder Charging	J1244 POI – Portage Tap – Atlantic 69 kV line	ATC	Base Case P11 P12 P13 P14 P21 P23	J1244
	J1244 POI – Winona 69 kV line	ATC	P12 P21	J1244

¹ The "No Load Loss Allowed" NERC TPL Planning Events refer to all the Planning Events in NERC TPL-001-4 Table 1 that the interruption of Firm Transmission Service and Non-Consequential Load Loss are not allowed.

² Per ATC's Planning Criteria, for Planning Events that allow Non-Consequential load loss, short duration exposure to potential post-contingency overload represents an acceptable risk assuming system adjustment within that short period of time can eliminate the exceedance and restore facility loading within Long-Term emergency ratings.

2.3.4 Network Upgrades Identified in ERIS Analysis

Based on the steady-state analyses, the worst loading of each facility under "No Load Loss Allowed" NERC TPL Planning Events that meets MISO Injection Constraint criteria is shown in Table 2.3.4-1. Potential Network Upgrades are also included. Good faith Cost Estimates of the ERIS thermal Network Upgrades identified in the steady-state analysis for the 2024 scenarios are listed in Table 2.3.4-2. Detailed cost allocations are provided in Section 9.

Table 2.3.4-1 – ERIIS Network Upgrades Identified to Address
MISO Steady-State Injection Constraints

Steady-State Injection Constraint	Responsible Generator(s)	Facility Owner	Study Case	Applicable Rating (MVA)	Worst Loading (%)	"No Load Loss Allowed" NERC TPL Planning Event	Network Upgrades
Dafter – Nine Mile 69 kV line	J1370	ATC	24SUM, Batteries Discharging	24	221.01		Dafter – Nine Mile (ESE_6957) 69 kV, Bifurcate
J1244 POI – Portage Tap 69 kV line	J1244	ATC	24SH, Batteries Discharging	37	110.24	ne	Winona - Atlantic (Winona69) 69kV, Reconductor; MTEP PID:4727
Portage Tap – Atlantic 69 kV line	J1244	ATC	24SH, Batteries Discharging	37	110.16		
J1244 POI – Winona 69 kV line	J1244	ATC	24SH, Batteries Charging	37	106.23		

Table 2.3.4-2 – ERIS Network Upgrades and Cost Estimates

Steady-State Injection Constraint	Facility Owner	Network Upgrade	Cost (\$) ^{1,2}
Dafter – Nine Mile 69 kV line	ATC	Dafter – Nine Mile (ESE_6957) 69 kV, Bifurcate	325,098
J1244 POI – Portage Tap 69 kV line	ATC	Winona - Atlantic (Winona69) 69kV, Reconductor; MTEP PID:4727	0
Portage Tap – Atlantic 69 kV line	ATC		
J1244 POI – Winona 69 kV line	ATC		

¹ All Network Upgrades were estimated on the generator ISD dollars.

² ATC Network Upgrades are estimated in ISD dollars with a 2.5% annual escalation rate and include a contingency based on a project risk generally between 5% – 20%.

2.3.5 Network Upgrade Alternatives Considered

The ERIS network upgrades identified in Table 2.3.4-2 are direct upgrades of the ERIS thermal constraint facilities to ATC design standards and considered as least-cost solutions. Therefore, no other alternatives were examined for those solutions.

2.3.6 Potential Operating Restrictions

The purpose of the study is to identify potential steady-state operating restrictions for study generators under prior outage conditions and raise awareness of these potential operating restrictions to customers. Real-time steady-state thermal constraints due to NERC Category P3/P6 events (N-1-1) will be mitigated in the day-ahead and real-time market through the MISO binding constraint and other operating procedures. The study was performed on study models with final ERIS Network Upgrades and NRIS Network Upgrades included. Based on ATC generator operating restriction study methodology as described in Appendix A, the worst potential steady-state operating restrictions for study generators were identified and summarized in Table 2.3.6-1. If multiple N-1-1 events resulted in the same MW restriction, only the worst N-1-1 event (highest loading % on the constraint) was listed. The full NERC Category P3/P6 steady-state study results were documented in Appendix F. Operating restrictions could occur if either of the contingent elements are out of service.

Table 2.3.6-1 – Worst Steady-State Operating Restrictions with ERIS and NRIS Network Upgrades Included

MISO Queue #	Model	Worst NERC Category P3/P6 (N-1-1) Event		Max MW Allowed
		1 st Contingency	2 nd Contingency	
J1214	-	-	-	-
J1244	SH_DIS, SH_CH	████████████████████	████████████████████ ████████████████████	30.70
J1251	-	-	-	-
J1253	SUM_DIS	████████████████████ ████	████████████████████ ████████████████████	0.00
J1304	-	-	-	-
J1305	-	-	-	-
J1316	-	-	-	-

J1326	-	-	-	-
J1345	SH_CH			-7 ¹
J1370	SUM_DIS			39.89
J1374	SH_DIS, SH_CH			0.00
J1377	-	-	-	-
J1410	-	-	-	-
J1411	-	-	-	-
J1460	-	-	-	-
J1483	SH_CH			0.00

¹J1345 would be restricted in charging mode in this scenario.

3.0 STABILITY ANALYSIS

Stability analysis in Section 3 was performed using TSAT V19 to evaluate the stability performance of the generating facilities in the DPP 2019 Cycle 1 Phase 2. Detailed study assumptions, criteria, and methodology are documented in Appendix A. For generators that require Electromagnetic transient (EMT) stability analysis using Power Systems Computer-Aided Design (PSCAD), results are documented in Section 6 of the report.

Stability plots consist of generator rotor angles, generator real power output, generator reactive power output, generator terminal voltages, and transmission bus voltages for each simulation. Simulations were performed with a 0.5 second flat start followed by the appropriate disturbance. Simulations were run for a 20-second duration.

3.1 Model Development

3.1.1 Study Case

A stability study case representing the 2024 shoulder load condition was developed by MISO and reviewed by ATC and Interconnection Customers. The model was developed and studied in TSAT v19. TSAT, like PSSE, is a root mean squared (RMS) based program that only models a network's positive sequence quantities. All units under study were dispatched to 100% of P_{MAX}. Additionally, the following local units were also dispatched to 100% of P_{MAX} to meet ATC Planning Criteria:

Table 3.1.1-1 – Local Unit Dispatch for Stability Analysis

Queue #	Nearby Units to Dispatch to P _{MAX}
J1214/J1326	Columbia, Rockgen, West Riverside
J1244	Mihm, Warden, Portage CT
J1251/J1252	Garden Corners Wind, J849, J928, J1183
J1253	Point Beach, Fox, South Fond du Lac, Columbia, Edgewater
J1304	West Riverside, Riverside, Rock River
J1305	Riverside, Rock River, Quilt Block, J947

J1316	Paris, J878, J1316, J1042
J1345	Glacier Hills, J1003
J1370	Dafter, US Hydro, ES Hydro
J1374	J855, J864, J870, J871, J947, J1000
J1377	Columbia, Rockgen, West Riverside, J818, J1154
J1410/J1411	Columbia, Rockgen, West Riverside, S Fond du Lac
J1460	Riverside, Rock River, University, J850, J1042
J1483	J855, J864, J870, J871, J947, J1000

3.1.2 Benchmark Case

The DPP 2019 Cycle 1 Phase 2 generating facilities were removed from the study case. The MISO Classic methodology, where generation was scaled in proportion to Pmax minus Pgen, was used for power balance.

Faults were simulated using the 2024 shoulder study case. If a transient stability criteria constraint was identified, the same disturbance was repeated in the benchmark case.

3.2 J1214/J1326 Stability Study

3.2.1 J1214/J1326 Stability Results

Complete fault definitions and stability results for J1214/J1326 can be found in Appendix G. Under all faults for J1214/J1326, the simulations show the system meeting all transient stability criteria.

3.2.2 Network Upgrades Identified in the Stability Study

Stability Network Upgrades are not required for the interconnection of the J1214/J1326 generating facilities.

3.2.3 Operating Restrictions Identified in the Stability Study

No stability operating restrictions were identified for the J1214/J1326 generating facilities.

3.2.4 PRC-024-2 Analyses

PRC-024 Analysis was completed for each generator using the methodology outlined in Appendix A section A.2.5 and the customer provided frequency and voltage trip settings. Results for J1214/J1326 are provided in tables below.

Table 3.2.4-1 – J1214/J1326 PRC-024-2 Voltage Protection Relay Settings Results

Relay Type	Voltage Threshold (p.u.)	Relay Time (Sec)	Ride Through PRC-024-2 "No Trip Zone"?
Low Voltage	0.5	1	Yes
Low Voltage	0.7	10	Yes
Low Voltage	0.88	20	Yes
High Voltage	1.1	12	Yes

High Voltage	1.2	0.05	Yes
High Voltage	1.3	0.033	Yes

Table 3.2.4-2 – J1214/J1326 PRC-024-2 Frequency Protection Relay Settings Results

Relay Type	Frequency Threshold (Hz)	Relay Time (Sec)	Minimum Required Relay Time at This Frequency (Sec)	Ride Through PRC-024-2 “No Trip Zone”?
Low Frequency	57	0.16	0	Yes
Low Frequency	58.5	300	33	Yes
High Frequency	60.5	605	601	Yes
High Frequency	62	0.16	0	Yes

3.3 J1244 Stability Study

3.3.1 J1244 Stability Results

Complete fault definitions and stability results for J1244 can be found in Appendix G. The following four P6 events showed unstable results for J1244:

- [REDACTED]
- [REDACTED]
- [REDACTED]
- [REDACTED]

Under all other faults for J1244, the simulations show the system meeting all transient stability criteria.

3.3.2 Network Upgrades Identified in the Stability Study

Stability Network Upgrades are not required for the interconnection of the J1244 generating facilities.

3.3.3 Operating Restrictions Identified in the Stability Study

The following stability operating restrictions were identified for the J1244 generating facilities.

Table 3.3.3-1 – J1244 Stability Operating Restrictions

Prior Outage	Faulted Element	Mihm Restriction	J1244 Restriction
[REDACTED]	[REDACTED]	N/A	4 MW
[REDACTED]	[REDACTED]	N/A	22 MW
[REDACTED]	[REDACTED]	39 MW	0 MW

39 MW

N/A

3.3.4 PRC-024-2 Analyses

PRC-024 Analysis was completed for each generator using the methodology outlined in Appendix A section A.2.5 and the customer provided frequency and voltage trip settings. Results for J1244 are provided in tables below.

Table 3.3.4-1 – J1244 PRC-024-2 Voltage Protection Relay Settings Results

Relay Type	Voltage Threshold (p.u.)	Relay Time (Sec)	Ride Through PRC-024-2 “No Trip Zone”?
Low Voltage	0.199	0.5	Yes
Low Voltage	0.499	1.1	Yes
Low Voltage	0.699	2.1	Yes
Low Voltage	0.799	4	Yes
Low Voltage	0.849	5	Yes
Low Voltage	0.899	180	Yes
High Voltage	1.101	180	Yes
High Voltage	1.121	10	Yes
High Voltage	1.201	1	Yes
High Voltage	1.251	0.5	Yes

Table 3.3.4-2 – J1244 PRC-024-2 Frequency Protection Relay Settings Results

Relay Type	Frequency Threshold (Hz)	Relay Time (Sec)	Minimum Required Relay Time at This Frequency (Sec)	Ride Through PRC-024-2 “No Trip Zone”?
Low Frequency	47.99	0.01	0	Yes
Low Frequency	55.19	0.1	0	Yes
Low Frequency	56.39	0.2	0	Yes
High Frequency	63.61	0.2	0	Yes
High Frequency	64.81	0.1	0	Yes
High Frequency	72.01	0.01	0	Yes

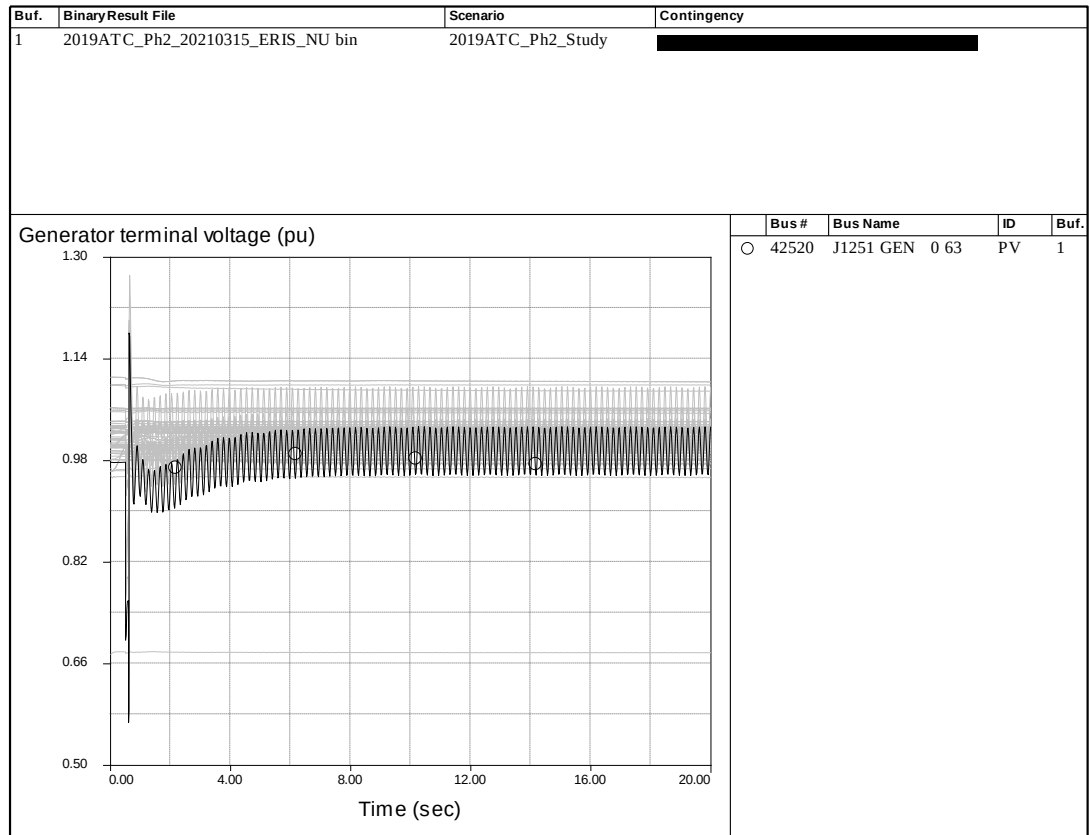
3.4 J1251 Stability Study

3.4.1 J1251 Stability Results

Complete fault definitions and stability results for J1251 can be found in Appendix G. [REDACTED]

[REDACTED]. This issue was determined to be a pre-existing modeling condition that will need to be restudied when ATC gets a TSAT user model for the Mackinac HVDC.

TSAT Mar 31, 2021 08:30:08



DSATools Output Analysis 20.0

Powertech Labs Inc.

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Figure 3.4.1-1 – J1251 Voltage Response to Contingency P71

Under all other faults for J1251, the simulations show the system meeting all transient stability criteria.

3.4.2 Network Upgrades Identified in the Stability Study

Stability Network Upgrades are not required for the interconnection of the J1251 generating facilities.

3.4.3 Operating Restrictions Identified in the Stability Study

No stability operating restrictions were identified for the J1251 generating facilities.

3.4.4 PRC-024-2 Analyses

PRC-024 Analysis was completed for each generator using the methodology outlined in Appendix A section A.2.5 and the customer provided frequency and voltage trip settings. Results for J1251 are provided in tables below.

Table 3.4.4-1 – J1251 PRC-024-2 Voltage Protection Relay Settings Results

Relay Type	Voltage Threshold (p.u.)	Relay Time (Sec)	Ride Through PRC-024-2 “No Trip Zone”?
Low Voltage	0.5	1	Yes
Low Voltage	0.7	10	Yes
Low Voltage	0.88	20	Yes
High Voltage	1.1	12	Yes
High Voltage	1.2	0.05	No
High Voltage	1.3	0.033	Yes

Table 3.4.4-2 – J1251 PRC-024-2 Frequency Protection Relay Settings Results

Relay Type	Frequency Threshold (Hz)	Relay Time (Sec)	Minimum Required Relay Time at This Frequency (Sec)	Ride Through PRC-024-2 “No Trip Zone”?
Low Frequency	57	0.16	0	Yes
Low Frequency	58.5	300	33	Yes
Low Frequency	60.5	605	601	Yes
High Frequency	62	0.16	0	Yes

3.5 J1253 Stability Study

3.5.1 J1253 Stability Results

Complete fault definitions and stability results for J1253 can be found in Appendix G. Under all faults for J1253, the simulations show the system meeting all transient stability criteria.

3.5.2 Network Upgrades Identified in the Stability Study

Stability Network Upgrades are not required for the interconnection of the J1253 generating facilities.

3.5.3 Operating Restrictions Identified in the Stability Study

No stability operating restrictions were identified for the J1253 generating facilities.

3.5.4 PRC-024-2 Analyses

PRC-024 Analysis was completed for each generator using the methodology outlined in Appendix A section A.2.5 and the customer provided frequency and voltage trip settings. Results for J1253 are provided in tables below.

Table 3.5.4-1 – J1253 PRC-024-2 Voltage Protection Relay Settings Results

Relay Type	Voltage Threshold (p.u.)	Relay Time (Sec)	Ride Through PRC-024-2 “No Trip Zone”?
Low Voltage	0.500	10.000	Yes
Low Voltage	0.700	21.000	Yes
Low Voltage	0.900	30.000	Yes
High Voltage	1.100	30.000	Yes

High Voltage	1.200	2.000	Yes
High Voltage	1.250	0.200	Yes

Table 3.5.4-2 – J1253 PRC-024-2 Frequency Protection Relay Settings Results

Relay Type	Frequency Threshold (Hz)	Relay Time (Sec)	Minimum Required Relay Time at This Frequency (Sec)	Ride Through PRC-024-2 “No Trip Zone”?
Low Frequency	56.5	6.0	0	Yes
High Frequency	63.5	6.0	0	Yes

3.6 J1304 Stability Study

3.6.1 J1304 Stability Results

Complete fault definitions and stability results for J1304 can be found in Appendix G. Under all faults for J1304, the simulations show the system meeting all transient stability criteria.

3.6.2 Network Upgrades Identified in the Stability Study

Stability Network Upgrades are not required for the interconnection of the J1304 generating facilities.

3.6.3 Operating Restrictions Identified in the Stability Study

No stability operating restrictions were identified for the J1304 generating facilities.

3.6.4 PRC-024-2 Analyses

PRC-024 Analysis was completed for each generator using the methodology outlined in Appendix A section A.2.5 and the customer provided frequency and voltage trip settings. Results for J1304 are provided in tables below.

Table 3.6.4-1 – J1304 PRC-024-2 Voltage Protection Relay Settings Results

Relay Type	Voltage Threshold (p.u.)	Relay Time (Sec)	Ride Through PRC-024-2 “No Trip Zone”?
Low Voltage	0.00	0.5	Yes
Low Voltage	0.60	1.0	Yes
Low Voltage	0.70	3.0	Yes
Low Voltage	0.89	6550.0	Yes
High Voltage	1.11	6550.0	Yes
High Voltage	1.20	0.5	Yes
High Voltage	1.30	0.01	Yes

Table 3.6.4-2 – J1304 PRC-024-2 Frequency Protection Relay Settings Results

Relay Type	Frequency Threshold (Hz)	Relay Time (Sec)	Minimum Required Relay Time at This Frequency (Sec)	Ride Through PRC-024-2 “No Trip Zone”?
Low Frequency	55.0	0.01	0	Yes
Low Frequency	57.0	10.00	0	Yes
Low Frequency	58.0	90.00	4	Yes

Low Frequency	59.3	6550.00	805	Yes
High Frequency	60.7	6550.00	307	Yes
High Frequency	62.0	90.00	0	Yes
High Frequency	63.0	5.00	0	Yes
High Frequency	64.0	0.01	0	Yes

3.7 J1305 Stability Study

3.7.1 J1305 Stability Results

Complete fault definitions and stability results for J1305 can be found in Appendix G. Under all faults for J1305, the simulations show the system meeting all transient stability criteria.

3.7.2 Network Upgrades Identified in the Stability Study

Stability Network Upgrades are not required for the interconnection of the J1305 generating facilities.

3.7.3 Operating Restrictions Identified in the Stability Study

No stability operating restrictions were identified for the J1305 generating facilities.

3.7.4 PRC-024-2 Analyses

PRC-024 Analysis was completed for each generator using the methodology outlined in Appendix A section A.2.5 and the customer provided frequency and voltage trip settings. Results for J1305 are provided in tables below.

Table 3.7.4-1 – J1305 PRC-024-2 Voltage Protection Relay Settings Results

Relay Type	Voltage Threshold (p.u.)	Relay Time (Sec)	Ride Through PRC-024-2 “No Trip Zone”?
Low Voltage	0.00	0.5	Yes
Low Voltage	0.60	1.0	Yes
Low Voltage	0.70	3.0	Yes
Low Voltage	0.89	6550.0	Yes
High Voltage	1.11	6550.0	Yes
High Voltage	1.20	0.5	Yes
High Voltage	1.30	0.01	Yes

Table 3.2.4-2 – J1305 PRC-024-2 Frequency Protection Relay Settings Results

Relay Type	Frequency Threshold (Hz)	Relay Time (Sec)	Minimum Required Relay Time at This Frequency (Sec)	Ride Through PRC-024-2 “No Trip Zone”?
Low Frequency	55.0	0.01	0	Yes
Low Frequency	57.0	10.00	0	Yes
Low Frequency	58.0	90.00	4	Yes
Low Frequency	59.3	6550.00	805	Yes
High Frequency	60.7	6550.00	307	Yes
High Frequency	62.0	90.00	0	Yes
High Frequency	63.0	5.00	0	Yes
High Frequency	64.0	0.01	0	Yes

3.8 J1316 Stability Study

3.8.1 J1316 Stability Results

Complete fault definitions and stability results for J1316 can be found in Appendix G. Under all faults for J1316, the simulations show the system meeting all transient stability criteria.

3.8.2 Network Upgrades Identified in the Stability Study

Stability Network Upgrades are not required for the interconnection of the J1316 generating facilities.

3.8.3 Operating Restrictions Identified in the Stability Study

No stability operating restrictions were identified for the J1316 generating facilities.

3.8.4 PRC-024-2 Analyses

PRC-024 Analysis was completed for each generator using the methodology outlined in Appendix A section A.2.5 and the customer provided frequency and voltage trip settings. Results for J1316 are provided in tables below.

Table 3.8.4-1 – J1316 PRC-024-2 Voltage Protection Relay Settings Results

Relay Type	Voltage Threshold (p.u.)	Relay Time (Sec)	Ride Through PRC-024-2 “No Trip Zone”?
Low Voltage	0.5	1	Yes
Low Voltage	0.7	10	Yes
Low Voltage	0.88	20	Yes
High Voltage	1.1	12	Yes
High Voltage	1.2	0.05	Yes
High Voltage	1.3	0.033	Yes

Table 3.8.4-2 – J1316 PRC-024-2 Frequency Protection Relay Settings Results

Relay Type	Frequency Threshold (Hz)	Relay Time (Sec)	Minimum Required Relay Time at This Frequency (Sec)	Ride Through PRC-024-2 “No Trip Zone”?
Low Frequency	57	0.16	0	Yes
Low Frequency	58.5	300	33	Yes
High Frequency	60.5	605	601	Yes
High Frequency	62	0.16	0	Yes

3.9 J1345 Stability Study

3.9.1 J1345 Stability Results

Complete fault definitions and stability results for J1345 can be found in Appendix G. Under all faults for J1345, the simulations show the system meeting all transient stability criteria.

3.9.2 Network Upgrades Identified in the Stability Study

Stability Network Upgrades are not required for the interconnection of the J1345 generating facilities.

3.9.3 Operating Restrictions Identified in the Stability Study

No stability operating restrictions were identified for the J1345 generating facilities.

3.9.4 PRC-024-2 Analyses

PRC-024 Analysis was completed for each generator using the methodology outlined in Appendix A section A.2.5 and the customer provided frequency and voltage trip settings. Results for J1345 are provided in tables below.

When PRC-024 analysis was performed with the battery in charging mode the plant would immediately ramp up from charging to 0 MW regardless of the voltage applied. A model will need to be provided that can maintain a charging scenario for J1345 and ride through the PRC-024 curves.

When the battery was in charging mode and the low voltage PRC-024 curve was applied J1345 entered momentary cessation. J1345 facilities will need to be set to not enter momentary cessation within the PRC-024 curves and a model that can represent this behavior will need to be provided.

Table 3.9.4-1 – J1345 PRC-024-2 Voltage Protection Relay Settings Results

Relay Type	Voltage Threshold (p.u.)	Relay Time (Sec)	Ride Through PRC-024-2 “No Trip Zone”?
Low Voltage	0.5	10	Yes
Low Voltage	0.7	21	Yes
Low Voltage	0.9	30	Yes
High Voltage	1.11	30	Yes
High Voltage	1.2	2	Yes
High Voltage	1.25	0.2	Yes

Table 3.9.4-2 – J1345 PRC-024-2 Frequency Protection Relay Settings Results

Relay Type	Frequency Threshold (Hz)	Relay Time (Sec)	Minimum Required Relay Time at This Frequency (Sec)	Ride Through PRC-024-2 “No Trip Zone”?
Low Frequency	56.5	6	0	Yes
High Frequency	63.5	6	0	Yes

3.10 J1370 Stability Study

3.10.1 J1370 Stability Results

Complete fault definitions and stability results for J1370 can be found in Appendix G. Under all faults for J1370, the simulations show the system meeting all transient stability criteria.

3.10.2 Network Upgrades Identified in the Stability Study

Stability Network Upgrades are not required for the interconnection of the J1370 generating facilities.

3.10.3 Operating Restrictions Identified in the Stability Study

The following stability operating restriction was identified for the J1370 generating facilities.

Table 3.10.3-1 – J1370 Stability Operating Restrictions

Prior Outage	Faulted Element	J1370 Restriction
		28 MW

3.10.4 PRC-024-2 Analyses

PRC-024 Analysis was completed for each generator using the methodology outlined in Appendix A section A.2.5 and the customer provided frequency and voltage trip settings. Results for J1370 are provided in tables below.

Table 3.2.4-1 – J1370 PRC-024-2 Voltage Protection Relay Settings Results

Relay Type	Voltage Threshold (p.u.)	Relay Time (Sec)	Ride Through PRC-024-2 “No Trip Zone”?
Low Voltage	0.650	0.383	No
Low Voltage	0.750	2.083	No
Low Voltage	0.900	3.083	No
High Voltage	1.100	1.083	Simulation Crashed
High Voltage	1.150	0.583	Simulation Crashed
High Voltage	1.175	0.283	Simulation Crashed
High Voltage	1.200	0.083	Simulation Crashed

Table 3.2.4-2 – J1370 PRC-024-2 Frequency Protection Relay Settings Results

Relay Type	Frequency Threshold (Hz)	Relay Time (Sec)	Minimum Required Relay Time at This Frequency (Sec)	Ride Through PRC-024-2 “No Trip Zone”?
Low Frequency	57.8	0.00005	0	Yes
Low Frequency	59.5	10.0000	1792	No
High Frequency	60.5	10.0000	601	No
High Frequency	61.8	0.00005	0	Yes

3.11 J1374 Stability Study

3.11.1 J1374 Stability Results

Complete fault definitions and stability results for J1374 can be found in Appendix G. Under all faults for J1374, the simulations show the system meeting all transient stability criteria.

3.11.2 Network Upgrades Identified in the Stability Study

Stability Network Upgrades are not required for the interconnection of the J1374 generating facilities.

3.11.3 Operating Restrictions Identified in the Stability Study

The following stability operating restriction was identified for the J1374 generating facilities.

Table 3.11.3-1 – J1374 Stability Operating Restrictions

Prior Outage	Faulted Element	J1374 Restriction
		0 MW

3.11.4 PRC-024-2 Analyses

PRC-024 Analysis was completed for each generator using the methodology outlined in Appendix A section A.2.5 and the customer provided frequency and voltage trip settings. Results for J1374 are provided in tables below.

Table 3.2.4-1 – J1374 PRC-024-2 Voltage Protection Relay Settings Results

Relay Type	Voltage Threshold (p.u.)	Relay Time (Sec)	Ride Through PRC-024-2 “No Trip Zone”?
Low Voltage	0.0001	0.450	Yes
Low Voltage	2.000	15.000	Yes
Low Voltage	0.400	2.600	Yes
Low Voltage	0.600	3.700	Yes
Low Voltage	0.800	4.800	Yes
Low Voltage	0.870	13.000	Yes
High Voltage	1.130	1800.000	Yes
High Voltage	1.160	60.000	Yes
High Voltage	1.250	2.000	Yes
High Voltage	1.360	0.150	Yes

Table 3.2.4-2 – J1374 PRC-024-2 Frequency Protection Relay Settings Results

Relay Type	Frequency Threshold (Hz)	Relay Time (Sec)	Minimum Required Relay Time at This Frequency (Sec)	Ride Through PRC-024-2 “No Trip Zone”?
Low Frequency	56.4	30.0	0	Yes
High Frequency	63.6	10.0	0	Yes

3.12 J1377 Stability Study

3.12.1 J1377 Stability Results

Complete fault definitions and stability results for J1377 can be found in Appendix G. Under all faults for J1377, the simulations show the system meeting all transient stability criteria.

3.12.2 Network Upgrades Identified in the Stability Study

Stability Network Upgrades are not required for the interconnection of the J1377 generating facilities.

3.12.3 Operating Restrictions Identified in the Stability Study

No stability operating restrictions were identified for the J1377 generating facilities.

3.12.4 PRC-024-2 Analyses

PRC-024 Analysis was completed for each generator using the methodology outlined in Appendix A section A.2.5 and the customer provided frequency and voltage trip settings. Results for J1377 are provided in tables below.

Table 3.2.4-1 – J1377 PRC-024-2 Voltage Protection Relay Settings Results

Relay Type	Voltage Threshold (p.u.)	Relay Time (Sec)	Ride Through PRC-024-2 “No Trip Zone”?
Low Voltage	0.40	10.00	Yes
Low Voltage	0.60	21.00	Yes
Low Voltage	0.70	60.00	Yes
Low Voltage	0.85	120.00	Yes
High Voltage	1.11	120.00	Yes
High Voltage	1.15	60.00	Yes
High Voltage	1.20	2.00	Yes
High Voltage	1.25	0.20	Yes

Table 3.2.4-2 – J1377 PRC-024-2 Frequency Protection Relay Settings Results

Relay Type	Frequency Threshold (Hz)	Relay Time (Sec)	Minimum Required Relay Time at This Frequency (Sec)	Ride Through PRC-024-2 “No Trip Zone”?
Low Frequency	57.8	30.00	0	Yes
Low Frequency	58.3	200.00	15	Yes
Low Frequency	58.9	2000.00	163	Yes
High Frequency	59.4	6550.00	1201	Yes
High Frequency	60.7	6550.00	307	Yes
High Frequency	61.0	2000.00	112	Yes
High Frequency	61.5	200.00	21	Yes

3.13 J1410/J1411 Stability Study

3.13.1 J1410/J1411 Stability Results

Complete fault definitions and stability results for J1410/J1411 can be found in Appendix G. Under all faults for J1410/J1411, the simulations show the system meeting all transient stability criteria.

3.13.2 Network Upgrades Identified in the Stability Study

Stability Network Upgrades are not required for the interconnection of the J1410/J1411 generating facilities.

3.13.3 Operating Restrictions Identified in the Stability Study

No stability operating restrictions were identified for the J1410/J1411 generating facilities.

3.13.4 PRC-024-2 Analyses

PRC-024 Analysis was completed for each generator using the methodology outlined in Appendix A section A.2.5 and the customer provided frequency and voltage trip settings. Results for J1410/J1411 are provided in tables below.

Table 3.2.4-1 – J1410/J1411 PRC-024-2 Voltage Protection Relay Settings Results

Relay Type	Voltage Threshold (p.u.)	Relay Time (Sec)	Ride Through PRC-024-2 “No Trip Zone”?
Low Voltage	0.5	1	Yes
Low Voltage	0.7	10	Yes
Low Voltage	0.88	20	Yes
High Voltage	1.1	12	Yes
High Voltage	1.2	0.05	Yes
High Voltage	1.3	0.033	Yes

Table 3.2.4-2 – J1410/J1411 PRC-024-2 Frequency Protection Relay Settings Results

Relay Type	Frequency Threshold (Hz)	Relay Time (Sec)	Minimum Required Relay Time at This Frequency (Sec)	Ride Through PRC-024-2 “No Trip Zone”?
Low Frequency	57	0.16	0	Yes
Low Frequency	58.5	300	33	Yes
High Frequency	60.5	605	601	Yes
High Frequency	62	0.16	0	Yes

3.14 J1460 Stability Study

3.14.1 J1460 Stability Results

Complete fault definitions and stability results for J1460 can be found in Appendix G. Under all faults for J1460, the simulations show the system meeting all transient stability criteria.

3.14.2 Network Upgrades Identified in the Stability Study

Stability Network Upgrades are not required for the interconnection of the J1460 generating facilities.

3.14.3 Operating Restrictions Identified in the Stability Study

No stability operating restrictions were identified for the J1460 generating facilities.

3.14.4 PRC-024-2 Analyses

PRC-024 Analysis was completed for each generator using the methodology outlined in Appendix A section A.2.5 and the customer provided frequency and voltage trip settings. Results for J1460 are provided in tables below.

Table 3.2.4-1 – J1460 PRC-024-2 Voltage Protection Relay Settings Results

Relay Type	Voltage Threshold (p.u.)	Relay Time (Sec)	Ride Through PRC-024-2 “No Trip Zone”?
Low Voltage	0.50	10.000	Yes
Low Voltage	0.70	21.000	Yes
Low Voltage	0.90	30.000	Yes
High Voltage	1.11	30.000	Yes
High Voltage	1.20	2.000	Yes
High Voltage	1.25	0.200	Yes

Table 3.2.4-2 – J1460 PRC-024-2 Frequency Protection Relay Settings Results

Relay Type	Frequency Threshold (Hz)	Relay Time (Sec)	Minimum Required Relay Time at This Frequency (Sec)	Ride Through PRC-024-2 “No Trip Zone”?
Low Frequency	56.5	6.0	0	Yes
High Frequency	63.5	6.0	0	Yes

3.15 J1483 Stability Study

3.15.1 J1483 Stability Results

Complete fault definitions and stability results for J1483 can be found in Appendix G. Under all faults for J1483, the simulations show the system meeting all transient stability criteria.

3.15.2 Network Upgrades Identified in the Stability Study

Stability Network Upgrades are not required for the interconnection of the J1483 generating facilities.

3.15.3 Operating Restrictions Identified in the Stability Study

No stability operating restrictions were identified for the J1483 generating facilities.

3.15.4 PRC-024-2 Analyses

PRC-024 Analysis was completed for each generator using the methodology outlined in Appendix A section A.2.5 and the customer provided frequency and voltage trip settings. Results for J1483 are provided in tables below.

Table 3.2.4-1 – J1483 PRC-024-2 Voltage Protection Relay Settings Results

Relay Type	Voltage Threshold (p.u.)	Relay Time (Sec)	Ride Through PRC-024-2 “No Trip Zone”?
Low Voltage	0.40	1.00	Yes

Low Voltage	6.00	1.70	Yes
Low Voltage	0.70	2.50	Yes
Low Voltage	0.75	3.00	Yes
Low Voltage	0.85	1.00	Yes
Low Voltage	0.90	600.00	Yes
High Voltage	1.1010	1.00	No
High Voltage	1.1500	5.00	Yes
High Voltage	1.1750	0.20	Yes
High Voltage	1.2000	0.10	Yes
High Voltage	1.3000	0.01	Yes

Table 3.2.4-2 – J1483 PRC-024-2 Frequency Protection Relay Settings Results

Relay Type	Frequency Threshold (Hz)	Relay Time (Sec)	Minimum Required Relay Time at This Frequency (Sec)	Ride Through PRC-024-2 “No Trip Zone”?
Low Frequency	55.0	0.26	0	Yes
Low Frequency	57.0	63.00	0	Yes
High Frequency	63.0	63.00	0	Yes
High Frequency	65.0	0.26	0	Yes

4.0 SHORT CIRCUIT ANALYSIS

Short Circuit analysis was performed in Phase 2 with the final ERIS and NRIS network upgrade projects included in ATC Protection CAPE (Computer-Aided Protection Engineering) models, which include all ATC-owned substations, all ATC local distribution customer owned substations, as well as limited buses into ATC neighboring systems (MP, ComEd, XCEL, DPC, ITC). The analysis was performed according to ATC short circuit analysis methodology as described in Appendix A. Short circuit analysis results for facilities under MISO Functional Control are described in this section. The short circuit impact to facilities that are not under MISO Functional Control are described in Section 7.

4.1 Short Circuit Analysis Results

Maximum and minimum fault duty was calculated at the POI for each generator request and results are summarized in Table 4.1-1, Table 4.1-2, Table 4.1-3 and Table 4.1-4.

Table 4.1-1 – Outage Assumptions for Minimum Fault Duty Calculations

MISO Queue #	Contingency
J1214	[REDACTED]
J1244	[REDACTED]
J1251	[REDACTED]
J1253	[REDACTED]
J1304	[REDACTED]
J1305	[REDACTED]
J1316	[REDACTED]
J1326	[REDACTED]
J1345	[REDACTED]
J1370	[REDACTED]
J1374	[REDACTED]
J1377	[REDACTED]
J1410	[REDACTED]
J1411	[REDACTED]
J1460	[REDACTED]
J1483	[REDACTED]

Table 4.1-2 – Maximum and Minimum Fault Duty at POIs

Location	Maximum Fault Duty (Amps)		Minimum Fault Duty (Amps)	
	Single-Phase	Three-Phase	Single-Phase	Three-Phase
J1214 POI without J1214	19,650	20,778	16,275	17,447
J1214 POI with J1214	21,014	21,452	17,649	18,121
J1244 POI without J1244	3,189	3,869	1,053	1,362
J1244 POI with J1244	4,618	4,435	2,318	1,926
J1251 POI without J1251	6,142	7,793	3,333	4,504
J1251 POI with J1251	7,795	8,275	4,963	4,985
J1253 POI without J1253	9,433	13,334	4,162	6,634
J1253 POI with J1253	10,676	13,787	5,496	7,086
J1304 POI without J1304	23,340	23,873	9,018	12,006
J1304 POI with J1304	24,235	24,235	10,281	12,367
J1305 POI without J1305	5,004	6,622	1,762	2,115
J1305 POI with J1305	5,934	6,894	2,496	2,388
J1316 POI without J1316	33,915	32,166	8,435	13,032
J1316 POI with J1316	34,526	32,445	9,668	13,309
J1326 POI without J1326	20,466	21,283	16,275	17,447
J1326 POI with J1326	21,014	21,452	16,835	17,616
J1345 POI without J1345	9,197	8,423	2,768	4,295
J1345 POI with J1345	10,025	8,738	4,157	4,600
J1370 POI without J1370	3,851	4,150	2,706	3,176
J1370 POI with J1370	5,731	6,025	4,619	5,049
J1374 POI without J1374	15,661	15,405	1,596	2,324
J1374 POI with J1374	16,571	15,746	2,788	2,664
J1377 POI without J1377	11,493	14,314	4,193	5,209
J1377 POI with J1377	12,088	14,511	4,766	5,406
J1410 POI without J1410	16,899	18,967	12,808	14,599
J1410 POI with J1410	19,045	19,640	14,988	15,272
J1411 POI without J1411	18,536	19,472	12,808	14,599
J1411 POI with J1411	19,045	19,640	13,418	14,767
J1460 POI without J1460	12,237	16,588	7,167	11,002
J1460 POI with J1460	16,720	17,661	11,846	12,071
J1483 POI without J1483	8,393	10,282	5,414	5,997
J1483 POI with J1483	9,272	10,671	5,609	6,387

Table 4.1-3 – Maximum Fault Thevenin Equivalent Data

Location	Positive Sequence (Ohm)	Negative Sequence (Ohm)	Zero Sequence (Ohm)	Three-Phase Thevenin X/R Ratio	Single-Phase Thevenin X/R Ratio	Three-Phase ANSI 37.010 X/R Ratio	Single-Phase ANSI 37.010 X/R Ratio
J1214 POI without J1214	0.59036 + j9.93888	0.70569 + j9.91463	1.43669 + j11.4788	16.8	11.5	22.5	19.1
J1214 POI with J1214	0.59036 + j9.93888	0.70569 + j9.91463	1.19642 + j10.3769	16.8	12.1	22.5	19.8
J1244 POI without J1244	1.99464 + j10.1015	2.10547 + j10.1396	3.75000 + j16.3977	5.1	4.7	7.4	6.0
J1244 POI with J1244	1.99464 + j10.1015	2.10547 + j10.1396	1.08116 + j8.96702	5.1	5.6	7.4	8.3
J1251 POI without J1251	1.68938 + j10.1575	1.70900 + j10.1446	4.73162 + j17.9806	6.0	4.7	6.9	5.2
J1251 POI with J1251	1.68938 + j10.1575	1.70900 + j10.1446	2.14717 + j11.9984	6.0	5.8	6.9	8.6
J1253 POI without J1253	1.21056 + j14.8895	1.25106 + j14.8727	7.09168 + j32.8631	12.3	6.6	14.6	7.9
J1253 POI with J1253	1.21090 + j14.8916	1.25141 + j14.8747	4.97676 + j27.6400	12.3	7.7	14.6	12.8
J1304 POI without J1304	0.25235 + j3.44598	0.29110 + j3.44328	0.58346 + j3.63821	13.7	9.3	19.9	11.5
J1304 POI with J1304	0.25237 + j3.44610	0.29112 + j3.44340	0.51223 + j3.41419	13.7	9.8	19.9	13.4
J1305 POI without J1305	3.19293 + j11.6007	3.21984 + j11.5992	6.76253 + j22.7156	3.6	3.5	3.7	4.0
J1305 POI with J1305	3.19293 + j11.6007	3.21984 + j11.5992	3.84539 + j17.4537	3.6	4.0	3.7	5.2
J1316 POI without J1316	0.17937 + j2.60340	0.20632 + j2.61306	0.15141 + j2.11531	14.5	13.7	24.2	22.9
J1316 POI with J1316	0.17937 + j2.60340	0.20632 + j2.61306	0.14471 + j2.05066	14.5	13.7	24.2	22.4
J1326 POI without J1326	0.59036 + j9.93888	0.70569 + j9.91463	1.30855 + j10.9170	16.8	11.8	22.5	19.6
J1326 POI with J1326	0.59036 + j9.93888	0.70569 + j9.91463	1.19642 + j10.3769	16.8	12.1	22.5	19.8
J1345 POI without J1345	1.28254 + j5.17856	1.28389 + j5.17981	0.54093 + j3.93104	4.0	4.6	4.7	5.5
J1345 POI with J1345	1.28254 + j5.17856	1.28389 + j5.17981	0.54093 + j3.93104	4.0	4.6	4.7	5.5
J1370 POI without J1370	1.93234 + j9.40283	1.92625 + j9.38776	2.24652 + j11.6383	4.9	5.0	8.0	7.7
J1370 POI with J1370	1.29560 + j6.48413	1.30860 + j6.50699	1.52444 + j7.45142	5.0	5.0	7.1	6.7
J1374 POI without J1374	0.49915 + j5.81389	0.50388 + j5.81348	0.47361 + j5.50114	11.6	11.6	18.4	18.3
J1374 POI with J1374	0.49915 + j5.81389	0.50388 + j5.81348	0.37909 + j4.92534	11.6	12.0	18.8	19.3
J1377 POI without J1377	0.81399 + j13.9820	0.90544 + j13.9599	4.29266 + j24.0427	17.2	8.6	19.8	12.6
J1377 POI with J1377	0.81402 + j13.9823	0.90548 + j13.9602	3.66256 + j22.2092	17.2	9.3	19.8	17.9
J1410 POI without J1410	0.78032 + j10.9544	0.98081 + j10.9406	2.49683 + j14.2377	14.0	8.5	16.3	11.4
J1410 POI with J1410	0.78032 + j10.9544	0.98081 + j10.9406	1.65408 + j11.4213	14.0	9.8	18.1	15.0

J1411 POI without J1411	0.78032 + j10.9544	0.98081 + j10.9406	1.81370 + j12.0188	14.0	9.5	18.1	14.8
J1411 POI with J1411	0.78032 + j10.9544	0.98081 + j10.9406	1.65408 + j11.4213	14.0	9.8	18.1	15.0
J1460 POI without J1460	0.61099 + j4.76408	0.64408 + j4.76306	1.86433 + j9.75466	7.8	6.2	8.8	7.6
J1460 POI with J1460	0.61101 + j4.76421	0.64412 + j4.76319	0.63586 + j5.57517	7.8	8.0	8.8	10.1
J1483 POI without J1483	1.90804 + j20.6185	1.94440 + j20.6150	3.61704 + j34.4856	10.8	10.1	13.0	15.5
J1483 POI with J1483	1.90804 + j20.6185	1.94440 + j20.6150	2.77151 + j29.6080	10.8	10.7	11.6	14.5

Table 4.1-4 – Minimum Fault Thevenin Equivalent Data

Location	Positive Sequence (Ohm)	Negative Sequence (Ohm)	Zero Sequence (Ohm)	Three-Phase Thevenin X/R Ratio	Single-Phase Thevenin X/R Ratio	Three-Phase ANSI 37.010 X/R Ratio	Single-Phase ANSI 37.010 X/R Ratio
J1214 POI without J1214	0.66505 + j11.5561	0.77919 + j11.5266	1.62873 + j14.0094	17.4	12.1	23.3	20.2
J1214 POI with J1214	0.66505 + j11.5561	0.77919 + j11.5266	1.30866 + j12.3998	17.4	12.9	23.3	20.9
J1244 POI without J1244	6.66532 + j28.4828	6.84582 + j28.7312	12.8801 + j53.1348	4.3	4.2	5.6	4.8
J1244 POI with J1244	6.66532 + j28.4828	6.84582 + j28.7312	0.88245 + j14.2836	4.3	5.0	5.6	6.4
J1251 POI without J1251	3.35465 + j17.3708	3.39473 + j17.3895	9.70730 + j35.0423	5.2	4.2	5.6	4.6
J1251 POI with J1251	3.35465 + j17.3708	3.39473 + j17.3895	2.59984 + j17.7189	5.2	5.6	5.6	7.4
J1253 POI without J1253	2.80438 + j29.8943	2.87854 + j29.8157	19.6524 + j81.6201	10.7	5.6	11.6	6.1
J1253 POI with J1253	2.80598 + j29.9027	2.88018 + j29.8241	8.93830 + j55.5465	10.7	7.9	11.6	13.4
J1304 POI without J1304	0.73236 + j6.64167	0.80008 + j6.63489	2.74647 + j12.9722	9.1	6.1	10.9	6.5
J1304 POI with J1304	0.73246 + j6.64212	0.80018 + j6.63534	1.78426 + j10.5368	9.1	7.2	10.9	10.4
J1305 POI without J1305	9.57673 + j36.4420	9.57975 + j36.4417	17.7967 + j57.6025	3.8	3.5	4.0	3.9
J1305 POI with J1305	9.57673 + j36.4420	9.57975 + j36.4417	5.28154 + j32.4161	3.8	4.3	4.0	5.2
J1316 POI without J1316	0.78498 + j6.06317	0.79147 + j6.06351	3.66561 + j15.7215	7.7	5.3	12.9	5.6
J1316 POI with J1316	0.78498 + j6.06317	0.79147 + j6.06351	248561 + j12.7962	7.7	6.1	8.2	8.4
J1326 POI without J1326	0.66505 + j11.5561	0.77919 + j11.5266	1.62873 + j14.0094	17.4	12.1	23.3	20.2
J1326 POI with J1326	0.66505 + j11.5561	0.77919 + j11.5266	1.45390 + j13.1317	17.4	12.5	23.3	20.4
J1345 POI without J1345	2.93069 + j8.79970	2.93194 + j8.80235	7.68195 + j23.3901	3.0	3.0	3.1	3.1
J1345 POI with J1345	2.93161 + j8.80093	2.93286 + j8.80358	2.38233 + j12.3737	3.0	3.6	3.1	4.1

J1370 POI without J1370	2.90041 + j12.2039	2.90032 + j12.2043	5.06534 + j18.3955	4.2	3.9	6.4	4.7
J1370 POI with J1370	1.68653 + j7.70715	1.71066 + j7.74936	2.45158 + j9.74548	4.6	4.3	5.9	4.9
J1374 POI without J1374	4.80664 + j33.9425	4.80692 + j33.9426	12.6280 + j80.1865	7.1	6.7	186.4	13.8
J1374 POI with J1374	4.80664 + j33.9425	4.80692 + j33.9426	1.69583 + j29.6987	7.1	8.6	141.2	68.0
J1377 POI without J1377	2.28335 + j38.1669	2.45702 + j38.1059	9.50309 + j65.5312	16.7	10.0	18.1	14.3
J1377 POI with J1377	2.28364 + j38.1693	2.45733 + j38.1083	6.35578 + j53.3792	16.7	11.7	18.1	21.4
J1410 POI without J1410	0.94752 + j13.6111	1.21697 + j13.5981	3.61295 + j19.0879	14.4	8.0	18.5	10.6
J1410 POI with J1410	0.94752 + j13.6111	1.21697 + j13.5981	2.09250 + j14.2834	14.4	9.7	18.5	15.2
J1411 POI without J1411	0.94752 + j13.6111	1.21697 + j13.5981	3.61295 + j19.0879	14.4	8.0	18.5	10.6
J1411 POI with J1411	0.94752 + j13.6111	1.21697 + j13.5981	3.07828 + j17.5328	14.4	8.5	18.5	12.3
J1460 POI without J1460	1.12586 + j7.15391	1.14597 + j7.15614	3.78805 + j18.4836	6.4	5.4	6.9	6.7
J1460 POI with J1460	1.12595 + j7.15420	1.14607 + j7.15643	0.69930 + j7.62330	6.4	7.4	6.9	8.7
J1483 POI without J1483	3.87369 + j37.0155	3.87581 + j37.0156	5.12273 + j48.9541	9.6	9.6	12.8	15.8
J1483 POI with J1483	3.87369 + j37.0155	3.87581 + j37.0156	3.55425 + j39.6733	9.6	10.1	10.2	13.0

ATC breaker duty analysis results for over duty breakers are summarized in Table 4.1-5 and Table 4.1-6. No over duty breakers were identified.

Table 4.1-5 – Over Duty Breakers Found in SLG Analysis

Breaker Name	Breaker Rating	Base Case SLG Margin	Study Case SLG Margin
None	N/A	N/A	N/A

Table 4.1-6 – Over Duty Breakers Found in 3PG Analysis

Breaker Name	Breaker Rating	Base Case 3PG Margin	Study Case 3PG Margin
None	N/A	N/A	N/A

Please note, no breaker duty analysis was performed for any Affected System equipment. Appendix N provides short circuit data for all substation buses 69 kV and above included in ATC protection CAPE models. Any non-ATC equipment requiring a change would be considered Affected System.

4.2 Network Upgrades Identified in the Short Circuit Study

Table 4.2-1 Short circuit Network Upgrades identified

Over Duty Breakers	Facility Owner	Network Upgrade	Cost (\$)
None	-	-	-

5.0 TRANSFORMER ENERGIZATION ANALYSIS

5.1 Transformer Energization Study Results

Transformer Energization Analysis were performed based on ATC transformer initial energization criteria as described in Appendix A. The 2024 shoulder model (batteries charging) was used with all identified ERIS and NRIS Network Upgrades included. The results are summarized in Table 5.1-1. No constraints were found.

Table 5.1-1 – Inrush Calculations Using Shoulder Model

Generation Project	Transformer(s)		PSSE POI	Fault	Vmin Inrush (pu)	
	Number	Windings	Bus #	Current (A)	Raw	Multiplier
J1214	2	2	73270	16608	0.955	1.098
J1244	2	2	42447	3014	0.818	0.941
J1251	1	2	42527	5759	0.818	0.940
J1253	1	3	42535	11736	0.917	1.054
J1304	1	2	43044	20989	0.963	1.108
J1305	1	2	43054	6048	0.908	1.044
J1316	1	3	699409	21775	0.970	1.116

J1326	1	2	73270	16608	0.976	1.123
J1345	1	2	698222	7143	0.848	0.975
J1370	1	2	699713	3207	0.694	0.798
J1374	1	2	88554	13277	0.946	1.088
J1377	1	2	43774	12423	0.950	1.092
J1410	2	2	999907	16071	0.954	1.097
J1411	1	2	999907	16071	0.976	1.122
J1460	2	2	698095	14966	0.915	1.052
J1483	1	2	693863	9156	0.945	1.086

6.0 PSCAD ANALYSIS

ATC performed a low short circuit strength screening analysis for all inverter-based resources in the DPP 2019 Cycle 1 East (ATC) Area Phase 1 which determined that the following requests that have proceeded to Phase 2 were connecting to relatively weaker portions of the ATC system that would require further electromagnetic transient (EMT) analysis in PSCAD:

- J1244
- J1251
- J1345
- J1304
- J1305
- J1374
- J1483
- J1460

PSCAD requires much more data and processing time than traditional steady state and dynamics software packages used in transmission planning such that system wide area models do not exist. To study the generators connecting to weaker portions of the ATC system, four different local models were produced including:

- Southwest Wisconsin (SW WI) Model
- J1003/J1345 Model
- J1244 Model
- J1251 Model

6.1 SW WI PSCAD Study

Full details of the Southwest Wisconsin PSCAD study can be found in Appendix L.

6.1.1 SW WI PSCAD Study Results

All events studied were found to be stable for all elements monitored. Full results of the Southwest Wisconsin PSCAD study are summarized in Table 6.2.1-1 below. Full detailed results for all PSCAD results are included in Appendix L – SW WI PSCAD Report.

J855, J1374, and J1000 reactive power support will need to be coordinated as MVAR looping was observed in the study. This was resolved by coordinating the droop control across all three plants.

Table 6.1.1-1 – SW WI PSCAD Results

Case #	Fault Description	Fault Type	Note #
1		LLLG	1
2		LG	1
3		LLLG	1
4		LG	1, 2
5		LLLG	1
6		LG	1
7		LLLG	1
8		LG	1
9		LLLG	1
10		LG	1
11		LLLG	1
12		LG	1
13		LLLG	1
14		LG	1
15		LLLG	1
16		LG	1
17		LLLG	1
18		LG	1
19		LLLG	1, 3
20		LG	1, 3
21		LG	1
22		LLLG	1
23		LG	1
24		LLLG	1
25		LG	1
26		LLLG	1
27		LG	1
28		LLLG	1
29		LG	1
30		LLLG	1
31		LG	1
32		LLLG	1
33		LG	1
34		LLLG	1
35		LG	1
36		LLLG	1
37		LG	1
38		LLLG	1
39		LG	1
40		LLLG	1
41		LG	1
42		LLLG	1
43		LG	1
44		LLLG	1
45		LG	1
46		LLLG	1
47		LG	1
48		LLLG	1
49		LG	1
50		LLLG	1
51		LG	1
52		LLLG	1
53		LG	1

Case #	Fault Description	Fault Type	Note #
54		LLLG	4
55		LG	5
56		LLLG	6
57		LG	6

Note #	Note
1	All DPP-2019 generator meet TPL-001-4 and ATC Dynamic Stability Planning Criteria
2	Quilt Block Wind Plant enters a 4 second period of high reactive power output with transient spikes in reactive power/voltage following the fault, after which the plant returns to normal operation
3	J855 reduces Q by >25 MVAR and J1000 increases Q by > 20 MVAR (neither Q settles by the end of a 20 second simulation). J1374 does not help control the voltage due to high droop.
4	J870/J871 shows oscillatory behavior post-fault, eventually tripping after 4s
5	J870/J871 shows oscillatory/unstable behavior post-fault but does not trip
6	J870/J871 and J864 do not meet damping criteria 20 seconds post-fault

6.1.2 Network Upgrades Identified in the SW WI PSCAD Study

No network upgrades were identified to mitigate constraints from the Phase 2 SW WI PSCAD study.

6.1.3 Operating Restrictions Identified in the SW WI PSCAD Study

No new operating restrictions were identified in this study. The known P6 constraints around Badger Hollow were confirmed.

6.2 J1003/J1345 PSCAD Study

Full details of the J1003/J1345 PSCAD study can be found in Appendix L.

6.2.1 J1003/J1345 PSCAD Study Results

All events studied were found to be stable for all elements monitored. Full results of the J1003/J1345 PSCAD study are summarized in Table 6.2.1-1 below. Full detailed results for all PSCAD results are included in Appendix L – J1003/J1345 PSCAD Report.

Table 6.2.1-1 – J1003/J1345 PSCAD Results

CTG #	NERC Category	Fault Description	Fault Type	J1345/J1003 Met TPL-001-4 and ATC Dynamic Stability Planning Criteria?	Note #
1	P1.3	[REDACTED]	LG	✓	
2			3PG	✓	
3	P4.5	[REDACTED]	LG	✓	
4		[REDACTED]	3PG	✓	
		[REDACTED]			
5	P6.2.1	[REDACTED]	LG	✓	
6		[REDACTED]	3PG	✓	
7	P6.2.1	[REDACTED]	LG	✓	
8		[REDACTED]	3PG	✓	

6.2.2 Network Upgrades Identified in the J1003/J1345 PSCAD Study

No network upgrades were identified to mitigate constraints from the Phase 2 J1003/J1345 PSCAD study.

6.2.3 Operating Restrictions Identified in the J1003/J1345 PSCAD Study

No operating restrictions were identified to mitigate constraints from the Phase 2 J1003/J1345 PSCAD study.

6.3 J1244 PSCAD Study

Full details of the J1244 PSCAD study can be found in Appendix L.

6.3.1 J1244 PSCAD Study Results

All N-1 events studied were found to be stable after the J1244 PSCAD model was tuned for low SCRs. There were three N-1-1 events where instability occurred. Results of the J1244 PSCAD study are summarized in Table 6.3.1-1 below. Full detailed results for all PSCAD results are included in Appendix L – J1244 PSCAD Report.

Table 6.3.1-1 – J1244 PSCAD Results

Scenario Description			Fault			Fault Recovery		Mihm Generation (MW)	J1244 Curtailed Maximum Generation for Stability Requirements (MW)	
Scenario	Base case	Prior outage	ID	Location	Type	Status with full power at J1244	Remark			
1	B1 (With sync. machines)	None	C1	██████████	LLLG	✓		Unit 1 to 3 -18.8 (Total - 56.4)	38.4 (No curtailment)	
2			C2	██████████	LG	✓				
3			C3	██████████	LLLG	✓				
4			C4	██████████	LG	✓				
5			C5	██████████	LLLG	✓				
6			C6	██████████	LG	✓				
7		PO1 - Winona 138/69 kV TF	C5	██████████	LLLG	✓		Unit 1 to 3 -18.8 (Total - 56.4)	3.8	
8			C6	██████████	LG	✓				
9		PO2 - Winona - Silver River 138 kV	C1	██████████	LLLG	X	1	Unit 1 to 3 -18.8 (Total - 56.4)	3.8	
10			C2	██████████	LG	X				
11-1	B3-1 (2 Mihm units, with sync machines)	PO3 - Silver River - North Lake 138 kV	C7	██████████	LLLG	X	2	Unit 1 to 2 -15.5 Unit 3 - OFF (Total – 31.0)	15.3	
12-1			C8	██████████	LG	X				
11-2	B3-2 (3 Mihm units, with sync machines)	PO3 - Silver River - North Lake 138 kV	C7	██████████	LLLG	X	1	Unit 1 to 3 -13.0 (Total - 39.0)	19.1	
12-2			C8	██████████	LG	X	2			
13	B2 (No sync. machines)	None	C1	██████████	LLLG	✓		Unit 1 to3 - OFF (Total - 0.0)	38.4 (No curtailment)	
14			C2	██████████	LG	✓				
15			C3	██████████	LLLG	✓				
16			C4	██████████	LG	✓				
17			C5	██████████	LLLG	✓				
18			C6	██████████	LG	✓				
19		PO1 - Winona 138/69 kV TF	C5	██████████	LLLG	✓		Unit 1 to3 - OFF (Total - 0.0)	3.8	
20			C6	██████████	LG	✓				
21		PO2 - Winona - Silver River 138	C1	██████████	LLLG	X	1	Unit 1 to3 - OFF (Total - 0.0)	3.8	
22			C2	██████████	LG	X				
23	B4 (1 Mihm unit, No sync machines)	PO3 - Silver River - North Lake 138 kV	C7	██████████	LLLG	✓		Unit 1 -18.8 Unit 2 to 3 - OFF (Total – 18.8)	38.4 (No curtailment)	
24			C8	██████████	LG	✓				
25	B1 (With sync. machines)	J1244 POI - Winona 69 kV	C9	██████████	LLLG	X	3	Unit 1 to 3 -18.8 (Total - 56.4)	0.0	
26			C10	██████████	LG	✓			38.4 (No curtailment)	
27	B2 (No sync. machines)		C9	██████████	LLLG	X		Unit 1 to3 - OFF (Total - 0.0)	0.0	
28			C10	██████████	LG	✓	3		38.4 (No curtailment)	
✓	Appropriate performance in FRT and recovery									
X	Failure in FRT and recovery (Stability criteria violation)									

Remark	Description
1	Peak-peak magnitude of oscillation exceeds 3% of rated values at the end of 20s.
2	Network is unable to maintain stability
3	Wind farm trips

6.3.2 Network Upgrades Identified in the J1244 PSCAD Study

No network upgrades were identified to mitigate constraints from the Phase 2 J1244 PSCAD study.

6.3.3 Operating Restrictions Identified in the J1244 PSCAD Study

For P6 stability constraints, operating restrictions may be required under prior outage conditions to avoid instability caused by the next event. The maximum

allowed real power outputs after prior outages were identified for the J1244 generating facility to mitigate P6 stability constraints. The dispatch of a local synchronous machine called Mihm has a very large impact on the strength and stability of J1244 for the outage of [REDACTED]

[REDACTED] A few different dispatches were examined and can be seen in Table 6.3.3-1 below.

These operating restrictions will be examined again prior to commercial operation and it is possible that these restrictions could vary based on changes in assumptions.

Table 6.3.3-1 – J1244 PSCAD Identified Operating Restrictions

TPL	Prior Outage	Faulted Element	To Maintain IBR Stability and Other Voltage and Oscillation Criteria	
			Maximum Real Power Output of J1244 (MW)	Maximum Real Power Output of Mihm (MW)
P6	[REDACTED]	[REDACTED]	3.8	Pmax
P6	[REDACTED]	[REDACTED]	0.0	Pmax
P6	[REDACTED]	[REDACTED]	38.4	1 unit (18.8 MW total)
P6	[REDACTED]	[REDACTED]	15.3	2 units x 15.5 MW each (31 MW total)
P6	[REDACTED]	[REDACTED]	19.1	3 units x 13 MW each (39 MW total)

6.4 J1251 PSCAD Study

Full details of the J1244 PSCAD study can be found in Appendix L.

6.4.1 J1251 PSCAD Study Results

All events studied were found to be stable for all elements monitored. Full results of the J1251 PSCAD study are summarized in Table 6.4.1-1 below. Full detailed results for all PSCAD results are included in Appendix L – J1251 PSCAD Report.

Table 6.4.1-1 – J1251 PSCAD Results

CTG #	NERC Category	Fault Description	Fault Type	J1251 Met TPL-001-4 and ATC Dynamic Stability Planning Criteria?		Note #
				Weak System	Normal System	
1	P1.2	[REDACTED]	LG	✓	✓	
2			3PG	✓	✓	
3	P1.3	[REDACTED]	LG	✓	✓	
4			3PG	✓	✓	
5	P6.2.1	[REDACTED]	LG	✓	✓	
6			3PG	✓	✓	
7	P7	[REDACTED]	LG	✓	✓	
8	P7	[REDACTED]	LG	✓	✓	
9	P7	[REDACTED]	LG	✓	✓	

✓ Plant successfully rides through
 ○ Plant ride through, some minor issues are observed
 ✗ Plant tripped or unstable

6.4.2 Network Upgrades Identified in the J1251 PSCAD Study

No network upgrades were identified to mitigate constraints from the Phase 2 J1251 PSCAD study.

6.4.3 Operating Restrictions Identified in the J1251 PSCAD Study

No operating restrictions were identified to mitigate constraints from the Phase 2 J1251 PSCAD study.

6.5 Summary of Operating Restriction Identified in the PSCAD Analysis

For NERC Category P6 stability constraints, operating restrictions may be required under prior outages conditions to avoid instability caused by the next event. A summary of all P6 operating restrictions found in the PSCAD analysis for study generators in DPP 2019 Cycle 1 East (ATC) Area Phase 2 is provided in Table 6.5-1 below. Maximum allowed real power outputs after prior outages were identified for generating facility to mitigate P6 stability constraints.

Table 6.5-1 – Summary of All PSCAD Identified Operating Restrictions

MISO Queue #	Pmax	Worst NERC Category P6 (N-1-1) Event		PSCAD Analysis Max MW Allowed
		1st Contingency	2nd Contingency	
J870/J871	300	[REDACTED]	[REDACTED]	150
J870/J871	300	[REDACTED]	[REDACTED]	260
J1244	38.4	[REDACTED]	[REDACTED]	3.8

J1244	38.4			0.0
J1244	38.4			15.3

7.0 AFFECTED SYSTEM STUDY

7.1 Wisconsin Power and Light Affected System Study Summary

Based on the Phase 2 short circuit study results, Wisconsin Power and Light (WPL) provided the following estimated upgrade scopes and associated cost estimates for mitigating increased fault current at Sunrise and Wilcox substations. Detailed cost allocation information is provided in Section 9.

Table 7.1-1 – WPL Affected System Upgrades

Bus	kV	Substation	Total Estimated Upgrade Cost	Current Limiting Fuses upgrade	Protection Relay Setting Changes	Grounding Improvements
809500	138	SUNRISE	\$112,000	\$101,000	\$ 11,000	Already included in ATC J1460 Interconnection Facility scope
914900	138	WILCOX	\$ 212,000	\$101,000	\$ 11,000	\$100,000

7.2 PJM Affected System Steady State Study Summary

PJM performed the PJM Affected System analysis and the report can be found in Appendix H. Two PJM Affected System Network Upgrades have cost allocations to the study generators in DPP 2019 Cycle 1 Phase 2 East (ATC) group. Detailed cost allocations are provided in Section 9.

The PJM Affected System Report includes Steady State and Short Circuit analysis only. PJM will retool after Decision Point II with MISO and PJM higher-queued system changes, project withdrawals, and MW reductions as well as results from stability analysis. This retool will be provided for review in Phase III of the DPP-2019 cycle.

8.0 DELIVERABILITY STUDY

Generator interconnection requests have to pass Generator Deliverability Study to be granted Network Resource Interconnection Services (NRIS). If the generator is determined as not fully deliverable, the customer can either choose to elect the amount of NRIS available without upgrades or build system upgrades that will make the generator fully deliverable. Generator Deliverability Study ensures that the Network Resources, on an aggregate basis, can meet the MISO aggregate load requirements during system peak condition without getting bottled up.

MISO Generator Deliverability Study methodology is described in MISO BPM-15.

8.1 Study Summary

The summary of MISO deliverability results based on the 2024 summer peak study model is shown in the following tables.

Table 8.1-1 below lists the deliverability results with ERIIS Network Upgrades included in the NRIS analysis. Minimum NR Deliverable is the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew. Maximum NR Deliverable is the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared deduction.

Table 8.1-1 – NRIS Analysis Summary

MISO Queue #	Area	NR Tested (MW)	Minimum NR Deliverable (MW)	Maximum NR Deliverable (MW)
J1214	ATC	300	0	0
J1244	ATC	38.4	11.22	24.81
J1251	ATC	100	100	100
J1253	ATC	100	0	46.78
J1304	ATC	65	0	0
J1305	ATC	50	0	0
J1316	ATC	50	0	0
J1326	ATC	75	0	0
J1345	ATC	25	0	0
J1370	ATC	50	31.27	31.27
J1374	ATC	67.2	0	58.55
J1377	ATC	98.6	0	0
J1410	ATC	300	0	0
J1411	ATC	75	0	0
J1460	ATC	200	0	0

Table 8.1-2 below lists all of the NRIS constraints from the deliverability study and the identified NRIS Network Upgrades. Both ERIIS Network Upgrades and NRIS Network Upgrades must be made for 100% NRIS, i.e. fully deliverable. Please note, if a NRIS Network Upgrade entirely or partially changes the scope of an ERIIS Network Upgrade, only the cost difference between the NRIS upgrade and the ERIIS upgrade will be eligible for NRIS Network Upgrade cost allocation. Detailed NRIS Network Upgrade cost allocation calculations are provided in Section 9.

Table 8.1-2 – Network Upgrades Needed to Address MISO Identified NRIS Steady-State Injection Constraints

NRIS Thermal Constraints	Required NRIS Network Upgrades	Mitigation Type	Responsible Generators	NRIS Network Upgrade Cost Estimates (\$) [a]	ERIS Network Upgrade (Identified for Same Constraint) Cost Estimate (\$) [b]	Previous Cycle ERIS/NRIS Network Upgrade (Identified for Same Constraint) Cost Estimate (\$) [c]	Cost Used for NRIS Network Upgrade Allocation (\$) [d] = [a] - [b] - [c]
699551 NEEVIN-WEC 138 699525 BD MRT2 138 1	Neevin – Butte Des Morts 138 kV (43021), Rebuild	NR	J1253, J1345, J1410, J1411	1,556,274	0	0	1,556,274
699512 UNVRSTY 138 699357 MUKWONGO 138 1	University – Mukwonago 138 kV (UNIG52), Rebuild	NR	J1304, J1305, J1460	31,814,057	0	0	31,814,057
698884 JEFRSN5 138 699375 CRWFSH R 138 1	Jefferson – Crawfish River 138 kV (6654), Rebuild	NR	J1214, J1326, J1377, J1410, J1411	3,147,000	0	0	3,147,000
699175 NFL 138 138 699677 AVIATION 138 1	North Fond Du Lac – Aviation 138 kV (G111), Rebuild	NR	J1345	12,443,358	0	0	12,443,358
699677 AVIATION 138 699663 PROGRESS AVE 138 1 699663 PROGRESS AVE 138 699673 EOD_BUS2 138 1	Aviation - Ellinwood 138 kV (X-50), Rebuild	NR	J1345	2,807,900	0	0	2,807,900
699749 NINEMILE 69.0 699475 KINCHOT 69.0 1 699475 KINCHOT 69.0 699776 TONE 69.0 1 699776 TONE 69.0 699725 RUDYRD 69.0 1 699725 RUDYRD 69.0 699751 PINERV 69.0 1 699749 NINEMILE 69.0 699751 PINERV 69.0 1	Nine Mile – Pine River 69 kV (6923/6921), Double Circuit Rebuild	NR	J1370	25,477,000	0	0	25,477,000
694106 ROR BUS1 138 699062 MRE 138 138 1	Rock River - Marine Tap 138 kV (X-23), Partial Reconductor	NR	J1304, J1305, J1460	1,098,413	0	0	1,098,413
88704 J870J871 POI 138 699115 WYV 138 138 1	Barneveld Military Ridge Interconnection Project (MTEP PID:18052)	NR	J1374	0	0	0	0
	Eden - Highland 138 kV (X-147), Reconductor	NR	J1374	3,570,774	0	0	3,570,774
694022 FOXRIVER B1 345 699359 N APPLETON 345 1	Fox River – North Appleton 345 kV (NAPL121), Uprate	NR	J1316	292,629	0	0	292,629
699085 NLG 138 138 699360 NLK GV T 138 1	North Lake Geneva – North Lake Geneva Tap 138 kV (6541), Uprate	NR	J1304, J1305, J1460	411,985	0	0	411,985
699360 NLK GV T 138 699267 BRLGTN1 138 1	North Lake Geneva Tap – Burlington 138 kV (6541), Reconductor	NR	J1304, J1305, J1460	4,309,017	0	3,963,594	345,423
698883 WHTWTR5 138 699516 BLUFFCRK 138 1 699516 BLUFFCRK 138 699488 SGR CK5 138 1	White Water – Bluff Creek – Sugar Creek 138 kV (WHIG53), Uprate	NR	J1304, J1305, J1460	399,588	0	0	399,588
699086 ELK 138 138 699085 NLG 138 138 1	Elkhorn – North Lake Geneva 138 kV (X-55), Uprate	NR	J1304, J1305, J1460	52,259	0	45,400	6,859
698995 MASS 69.0 698997 BRUCE CR 69.0 1 698997 BRUCE CR 69.0 698999 WATERSM 69.0 1	Mass – Bruce Crossing – Watersmeet 69 kV (6530), Uprate	NR	J1244	964,627	0	0	964,627
699157 COL 345 345 694069 SFL 345 B1 345 1	Columbia – South Fond du Lac 345 kV (W-5), Uprate	NR	J1214, J1304, J1305, J1326, J1374, J1377, J1410, J1411	578,748	0	0	578,748

699340 JEFRSN4	138	698884 JEFRSN5	138	Z	Jefferson 138 kV, Replace Switches	NR	J1214, J1326, J1377	194,675	0	0	194,675
699375 CRWFSH R	138	699283 CONCRD 4	138	1	Crawfish River – Concord 138 kV (9043), Uprate	NR	J1214, J1326, J1377, J1410, J1411	2,084,375	0	902,919	1,181,456
89864 J986 POI	138	699241 POE 138	138	1	J986 POI – Port Edwards (X-11) 138 kV, Uprate	NR	J1345	4,652,851	0	584,677	4,068,174

8.2 Per Project Summary

8.2.1 J1214

J1214 Deliverable (NRIS) Amount in * Case: 0 MW (Conditional on ERS upgrades and case assumptions)					300 MW Requested								
Next Constraint for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)					Single Deduction	Minimum Level of Service Attainable (MW) *This would be the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew	Shared Deduction	Maximum level of service Attainable (MW) *This would be the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared	Max DF	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project	Total NRIS Cost of Upgrade	Notes
698884 JEFRSN5	138	699375 CRWFSH R	138	1	1723.16	0	336.68	0	0.062	J1214, J1326, J1377, J1410, J1411	1,229,848	3,147,000	
699375 CRWFSH R	138	699283 CONCRD 4	138	1	1434.62	0	280.3	19.7	0.062	J1214, J1326, J1377, J1410, J1411	461,713	1,181,456	
699340 JEFRSN4	138	698884 JEFRSN5	138	Z	751.78	0	245.36	54.64	0.063	J1214, J1326, J1377	127,084	194,675	
699157 COL 345	345	694069 SFL 345 B1	345	1	306.52	0	38.6	261.4	0.1164	J1214, J1304, J1305, J1326, J174, J1377, J1410, J1411	145,729	578,748	
						300		300					

8.2.2 J1244

J1244 Deliverable (NRIS) Amount in 2023 Case: 28.08 MW (Conditional on ERS upgrades and case assumptions)					38.2 MW Requested								
Next Constraint for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)					Single Deduction	Minimum Level of Service Attainable (MW) *This would be the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew	Shared Deduction	Maximum level of service Attainable (MW) *This would be the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared	Max DF	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project	Total NRIS Cost of Upgrade	Notes
698995 MASS	69.0	698997 BRUCE CR	69.0	1	48.72	0	24.36	13.84	0.1289	J1244	\$964,627	\$964,627	
698997 BRUCE CR	69.0	698999 WATERSM	69.0	1	7.34	30.86	3.68	34.52	0.1289	J1244	Cost included in above constraint	Cost included in above constraint	
						38.4		38.4					

8.2.3 J1251

J1251 Deliverable (NRIS) Amount in 2023 Case: 100 MW (Conditional on ERIIS upgrades and case assumptions)	100 MW Requeste								
Next Constraint for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Single Deductio n	Minimum Level of Service Attainable (MW) *This would be the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew	Shared Deductio n	Maximum level of service Attainable (MW) *This would be the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared	Max DF	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project	Total NRIS Cost of Upgrade	Notes
	100		100						

8.2.4 J1253

J1253 Deliverable (NRIS) Amount in * Case: 0 MW (Conditional on ERIIS upgrades and case assumptions)	100 MW Requeste								
Next Constraint for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Single Deductio n	Minimum Level of Service Attainable (MW) *This would be the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew	Shared Deductio n	Maximum level of service Attainable (MW) *This would be the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared	Max DF	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project	Total NRIS Cost of Upgrade	Notes
699551 NEEVIN-WEC 138 699525 BD MRT2 138 1 123.5		0	53.22	46.78	0.163	J1345, J1353, J1410, J1411	\$670,754	\$1,556,274	
		100		100					

8.2.5 J1304

J1304 Deliverable (NRIS) Amount in * Case: 0 MW (Conditional on ERIIS upgrades and case assumptions)	65 MW Requested								
Next Constraint for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Single Deduction	Minimum Level of Service Attainable (MW) *This would be the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew	Shared Deduction	Maximum level of service Attainable (MW) *This would be the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared	Max DF	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project	Total NRIS Cost of Upgrade	Notes
699512 UNVRSTY 138 699357 MUKWONGO 138 1 667.22		0	88.73	0	0.064	J1304, J1305, J1460	\$3,925,855	\$31,814,057	
699360 NLK GV T 138 699267 BRLGTN1 138 1 462.78		0	75.37	0	0.0967	J1304, J1305, J1460	59,931	345,423	
699157 COL 345 345 694069 SFL 345 B1 345 1 304.2		0	8.36	56.64	0.0587	J1214, J1304, J1305, J1326, J174, J1377, J1410, J1411	15,916	578,748	
699085 NLG 138 138 699360 NLK GV T 138 1 189.91		0	37.93	27.07	0.0912	J1304, J1305, J1460	82,933	411,985	
694106 ROR BUS1 138 699062 MRE 138 138 1 87.2		0	10.07	54.93	0.0722	J1304, J1305, J1460	126,757	1,098,413	
699516 BLUFFCRK 138 699488 SGR CK5 138 1 135.39		0	13	52	0.0573	J1304, J1305, J1460	38,360	399,588	
698883 WHTWTR5 138 699516 BLUFFCRK 138 1 72.97		0	7.01	57.99	0.0573	J1304, J1305, J1460	Cost included in above constraint	Cost included in above constraint	
699086 ELK 138 138 699085 NLG 138 138 1 11.44		53.56	2.26	62.74	0.1292	J1304 ,J1305, J1460	1,355	6,859	
		65		65					

8.2.6 J1305

J1305 Deliverable (NRIS) Amount in * Case: 0 MW (Conditional on ERIIS upgrades and case assumptions)					49.9 MW Requested									
Next Constraint for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)					Single Deduction	Minimum Level of Service Attainable (MW) *This would be the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew	Shared Deduction	Maximum level of service Attainable (MW) *This would be the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared	Max DF	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project	Total NRIS Cost of Upgrade	Notes	
699512	UNVRSTY	138	699357	MUKWONGO	138 1	714.65	0	68.12	0	0.0597	J1304, J1305, J1460	\$2,850,540	\$31,814,057	
699360	NLK GV T	138	699267	BRLGTN1	138 1	498.03	0	57.86	0	0.0898	J1304, J1305, J1460	42,763	345,423	
699085	NLG 138	138	699360	NLK GV T	138 1	207.73	0	29.12	20.78	0.0834	J1214, J1304, J1305, J1326, J174, J1377, J1410, J1411	57,596	411,985	
699157	COL 345	345	694069	SFL 345 B1	345 1	274.53	0	6.42	43.48	0.0836	J1304, J1305, J1460	13,543	578,748	
699516	BLUFFCRK	138	699488	SGR CK5	138 1	138.74	0	9.98	39.92	0.0559	J1304, J1305, J1460	28,770	399,588	
698883	WHTWTR5	138	699516	BLUFFCRK	138 1	74.77	0	5.38	44.52	0.0559	J1304, J1305, J1460	Cost included in above constraint	Cost included in above constraint	
694106	ROR BUS1	138	699062	MRE 138	138 1	47.37	2.53	7.73	42.17	0.1329	J1304, J1305, J1460	179,151	1,098,413	
699086	ELK 138	138	699085	NLG 138	138 1	12.45	37.45	1.45	48.45	0.1187	J1304, J1305, J1460	955	6,859	
						49.9			49.9					

8.2.7 J1316

J1316 Deliverable (NRIS) Amount in * Case: 0 MW (Conditional on ERIIS upgrades and case assumptions)					50 MW Requested									
Next Constraint for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)					Single Deduction	Minimum Level of Service Attainable (MW) *This would be the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew	Shared Deduction	Maximum level of service Attainable (MW) *This would be the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared	Max DF	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project	Total NRIS Cost of Upgrade	Notes	
694022	FOX RIVER B1	345	699359	N APPLETON	345 1	279.4	0	279.4	0	0.0594	J1316	\$292,629	\$292,629	
						50			50					

8.2.8 J1326

J1326 Deliverable (NRIS) Amount in * Case: 0 MW (Conditional on ERIIS upgrades and case assumptions)					75 MW Requested									
Next Constraint for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)					Single Deduction	Minimum Level of Service Attainable (MW) *This would be the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew	Shared Deduction	Maximum level of service Attainable (MW) *This would be the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared	Max DF	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project	Total NRIS Cost of Upgrade	Notes	
698884	JEFRSN5	138	699375	CRWF5H R	138 1	861.58	0	84.17	0	0.062	J1214, J1326, J1377, J1410, J1411	\$307,147	\$3,147,000	
699375	CRWF5H R	138	699283	CONCRD 4	138 1	717.19	0	70.06	4.94	0.062	J1214, J1326, J1377, J1410, J1411	115,310	1,181,456	
699340	JEFRSN4	138	698884	JEFRSN5	138 2	375.89	0	61.34	13.66	0.0628	J1214, J1326, J1377	31,771	194,675	
699157	COL 345	345	694069	SFL 345 B1	345 1	153.26	0	9.65	65.35	0.1164	J1214, J1304, J1305, J1326, J1374, J1377, J1410, J1411	36,461	578,748	
						75			75					

8.2.9 J1345

J1345 Deliverable (NRIS) Amount in * Case: 0 MW (Conditional on ERIIS upgrades and case assumptions)					25 MW Requested									
Next Constraint for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)					Single Deduction	Minimum Level of Service Attainable (MW) *This would be the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew	Shared Deduction	Maximum level of service Attainable (MW) *This would be the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared	Max DF	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project	Total NRIS Cost of Upgrade	Notes	
699175	NFL 138	138	699677	AVIATION 138 1	912.72	0	912.72	0	0.0525	J1345	\$12,443,358	\$12,443,358		
699677	AVIATION	138	699663	PROGRESS AVE 138 1	673.58	0	673.58	0	0.0936	J1345	2,807,900	2,807,900		
89864	J986 POI	138	699241	POE 138 1 C	508.51	0	508.51	0	0.0513	J1345	4,068,174	4,068,174		
699551	NEEVIN-WEC	138	699525	BD MRT2 138 1	244.87	0	13.31	0	0.082	J1345, J1253, J1410, J1411	84,661	1,556,274		
699663	PROGRESS AVE	138	699673	EOD_BUS2 138 1	116.98	0	116.98	0	0.0936	J1345	Cost Included in 2nd constraint from the top	Cost Included in 2nd constraint from the top		
						25		25						

8.2.10 J1370

J1370 Deliverable (NRIS) Amount in * Case: 31.27 MW (Conditional on ERIIS upgrades and case assumptions)					50 MW Requested									
Next Constraint for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)					Single Deduction	Minimum Level of Service Attainable (MW) *This would be the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew	Shared Deduction	Maximum level of service Attainable (MW) *This would be the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared	Max DF	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project	Total NRIS Cost of Upgrade	Notes	
699749	NINEMILE	69.0	699475	KINCHOT 69.0 1	18.73	31.27	18.73	31.27	0.7197	J1370	\$25,477,000	\$25,477,000		
699749	NINEMILE	69.0	699751	PINERV 69.0 1	13.8	36.2	13.8	35.13	0.7319	J1370				
699475	KINCHOT	69.0	699776	TONE 69.0 1	7.43	42.57	7.43	42.57	0.7197	J1370				
699776	TONE	69.0	699725	RUDYRD 69.0 1	6.29	43.71	6.29	43.71	0.7197	J1370				
699725	RUDYRD	69.0	699751	PINERV 69.0 1	4.58	45.42	4.58	45.42	0.7197	J1370				
						50		50						

8.2.11 J1374

J1374 Deliverable (NRIS) Amount in * Case: 0 MW (Conditional on ERIIS upgrades and case assumptions)					67.2 MW Requested									
Next Constraint for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)					Single Deduction	Minimum Level of Service Attainable (MW) *This would be the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew	Shared Deduction	Maximum level of service Attainable (MW) *This would be the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared	Max DF	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project	Total NRIS Cost of Upgrade	Notes	
699157	COL 345	345	694069	SFL 345 B1 345 1	183.26	0	8.65	58.55	0.0974	J1214, J1304, J1305, J1326, J174, J1377, J1410, J1411	\$27,317	\$578,748		
88704	J870J871 POI	138	699115	WYV 138 1 C	17.72	49.48	17.72	49.48	0.1563	J1374	3,570,774	3,570,774		
						67.2		67.2						

8.2.12 J1377

J1377 Deliverable (NRIS) Amount in * Case: 0 MW (Conditional on ERIIS upgrades and case assumptions)		98.6 MW Requested									
Next Constraint for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)		Single Deduction	Minimum Level of Service Attainable (MW) *This would be the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew	Shared Deduction	Maximum level of service Attainable (MW) *This would be the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared	Max DF	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project	Total NRIS Cost of Upgrade	Notes	
698884	JEFRSN5 138 699375 CRWFSH R 138 1	1013.34	0	110.65	0	0.0527	J1214, J1326, J1377, J1410, J1411	\$343,652	\$3,147,000		
699375	CRWFSH R 138 699283 CONCRD 4 138 1	843.65	0	92.12	6.48	0.0527	J1214, J1326, J1377, J1410, J1411	129,015	1,181,456		
699340	JEFRSN4 138 698884 JEFRSN5 138 2	438.16	0	80.64	17.96	0.0539	J1214, J1326, J1377	35,820	194,675		
699157	COL 345 345 694069 SFL 345 B1 345 1	179.54	0	12.69	85.91	0.0994	J1214, J1304, J1305, J1326, J174, J1377, J1410, J1411	40,917	578,748		
			98.6		98.6						

8.2.13 J1410

J1410 Deliverable (NRIS) Amount in * Case: 0 MW (Conditional on ERIIS upgrades and case assumptions)		300 MW Requested									
Next Constraint for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)		Single Deduction	Minimum Level of Service Attainable (MW) *This would be the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew	Shared Deduction	Maximum level of service Attainable (MW) *This would be the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared	Max DF	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project	Total NRIS Cost of Upgrade	Notes	
698884	JEFRSN5 138 699375 CRWFSH R 138 1	2091.92	0	336.68	0	0.0511	J1214, J1326, J1377, J1410, J1411	\$1,013,334	\$3,147,000		
699375	CRWFSH R 138 699283 CONCRD 4 138 1	1741.62	0	280.3	19.7	0.0511	J1214, J1326, J1377, J1410, J1411	380,429	1,181,456		
699551	NEEVIN-WEC 138 699525 BD MRT2 138 1	775.5	0	159.66	140.34	0.0518	J1253, J1345, J1410, J1411	640,718	1,556,274		
699157	COL 345 345 694069 SFL 345 B1 345 1	186.82	113.18	38.6	261.4	0.191	J1214, J1304, J1305, J1326, J174, J1377, J1410, J1411	239,081	578,748		
			300		300						

8.2.14 J1411

J1411 Deliverable (NRIS) Amount in * Case: 0 MW (Conditional on ERIIS upgrades and case assumptions)		75 MW Requested									
Next Constraint for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)		Single Deduction	Minimum Level of Service Attainable (MW) *This would be the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew	Shared Deduction	Maximum level of service Attainable (MW) *This would be the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared	Max DF	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project	Total NRIS Cost of Upgrade	Notes	
698884	JEFRSN5 138 699375 CRWFSH R 138 1	1045.96	0	84.17	0	0.0511	J1214, J1326, J1377, J1410, J1411	\$253,019	\$3,147,000		
699375	CRWFSH R 138 699283 CONCRD 4 138 1	870.81	0	70.07	4.93	0.0511	J1214, J1326, J1377, J1410, J1411	94,989	1,181,456		
699551	NEEVIN-WEC 138 699525 BD MRT2 138 1	387.75	0	39.92	35.08	0.0518	J1253, J1345, J1410, J1411	160,141	1,556,274		
699157	COL 345 345 694069 SFL 345 B1 345 1	93.41	0	9.65	65.35	0.191	J1214, J1304, J1305, J1326, J174, J1377, J1410, J1411	59,785	578,748		
			75		75						

8.2.15 J1460

J1460 Deliverable (NRIS) Amount in * Case: 0 MW (Conditional on ERIIS upgrades and case assumptions)					200 MW Requested									
Next Constraint for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)					Single Deduction	Minimum Level of Service Attainable (MW) *This would be the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew	Shared Deduction	Maximum level of service Attainable (MW) *This would be the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared	Max DF	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project	Total NRIS Cost of Upgrade	Notes	
699512	UNVRSTY	138	699357	MUKWONGO	138 1	707.6	0	273.02	0	0.1206	J1304, J1305, J1460	\$25,037,663	\$31,814,057	
699360	NLK GV T	138	699267	BRLGTN1	138 1	643.38	0	231.92	0	0.1391	J1304, J1305, J1460	242,729	345,423	
699085	NLG 138	138	699360	NLK GV T	138 1	353.58	0	116.7	83.3	0.098	J1214, J1304, J1305, J1326, J174, J1377, J1410, J1411	271,416	411,985	
699516	BLUFFCRK	138	699488	SGR CK5	138 1	96.18	103.82	40.02	159.98	0.1612	J1304, J1305, J1460	332,457	399,588	
694106	ROR BUS1	138	699062	MRE 138	138 1	85.9	114.1	30.98	169.02	0.1466	J1304, J1305, J1460	792,505	1,098,413	
698883	WHTWTR5	138	699516	BLUFFCRK	138 1	51.84	148.16	21.56	178.44	0.1612	J1304, J1305, J1460	Cost included in 4th constraint from the top	Cost included in 4th constraint from the top	
699086	ELK 138	138	699085	NLG 138	138 1	20.98	179.02	6.98	193.02	0.1409	J1304, J1305, J1460	4,548	6,859	
						200			200					

8.3 Network Upgrade Alternatives Considered

All NRIS network upgrades identified in Table 8.1-2 are direct upgrades of the constraint facilities to ATC design standards and considered as least-cost solutions at this point. Therefore, no other alternatives were examined.

9.0 COST ALLOCATION AND TRANSMISSION OWNER SELF-FUND ELECTION

The cost allocation of Network Upgrades for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of Energy Resource Interconnection services or Network Resource Interconnection services as of the System Impact Study report date. The cost estimate for each network upgrade was provided by the corresponding transmission owning company.

Per MISO tariff, Transmission Owners must provide a non-binding indication of whether or not they will self-fund any Network Upgrades identified in the System Impact Study analysis. The Self-Funding elections are shown in sections 9.1, 9.2, and 9.3.

9.1 Interconnection Facilities Proposed for DPP 2019 Cycle 1 Phase 2 Projects

For DPP 2019 Cycle 1 Phase 2 East (ATC) group, the total costs of interconnection facilities are summarized in Table 9.1-1.

Table 9.1-1 – Interconnection Facilities for DPP 2019 Cycle 1 Phase 2 Projects

MISO Queue #	Interconnection Facilities (\$)	
	TO Network Upgrades	Self-Fund
J1214	1,359,849	No
J1244	6,841,936	No
J1251	7,623,741	No
J1253	11,991,921	No
J1304	8,749,504	No
J1305	7,766,049	No
J1316	0	NA
J1326	1,359,849	No
J1345	0	NA
J1370	0	NA
J1374	1,201,495	No
J1377	13,937,272	No
J1410	4,436,635	No
J1411	4,436,635	No
J1460	4,065,157	No
J1483	1,613,040	No

9.2 ERIS Network Upgrades Proposed for DPP 2019 Cycle 1 Phase 2 Projects

Network upgrades for Energy Resource Interconnection Service (ERIS) were identified in the ERIS analysis. The ERIS network upgrades include thermal network upgrades and voltage support network upgrades identified in the steady-state analysis, stability network upgrades identified in the dynamic stability analysis, short circuit network upgrades identified in the short circuit analysis. EMT stability network upgrades identified in the PSCAD analysis and network upgrades identified in PJM Affected System analysis. For DPP 2019 Cycle 1 Phase 2 East (ATC) group, the total costs of ERIS network upgrades for the 2024 scenario are summarized in Tables 9.2-1 - Table 9.2-7.

Table 9.2-1 – ERIS Network Upgrades Identified in Steady State Analysis

Network Upgrade	Facility Owner	ISD (For Cost Estimate Only)	Cost (\$) ^{1,2}	Self-Fund
Dafter – Nine Mile (ESE_6957) 69 kV, Bifurcate	ATC	3/01/2021	325,098	No
Winona - Atlantic (Winona69) 69kV, Reconductor MTEP PID:4727	ATC	12/01/2024	0	NA

¹ All Network Upgrades were estimated on the earliest ISD dollars of responsible generator.

² ATC Network Upgrades are estimated in ISD dollars with a 2.5% annual escalation rate and include a contingency based on a project risk generally between 5% – 20%.

Table 9.2-2 – ERIS Network Upgrades Identified in Dynamic Stability Analysis

Network Upgrade	Facility Owner	ISD (For Cost Estimate Only)	Cost (\$)	Self-Fund
None	-	-	-	-

Table 9.2-3 – ERIS Network Upgrades Identified in Short Circuit Analysis

Network Upgrade	Facility Owner	ISD (For Cost Estimate Only)	Cost (\$)	Self-Fund
None	-	-	-	-

Table 9.2-4 – ERIS Network Upgrades Identified in PSCAD Analysis

Network Upgrade	Facility Owner	ISD (For Cost Estimate Only)	Cost (\$)	Self-Fund
None	-	-	-	-

Table 9.2-5 – ERIS Alliant Energy Affected System Upgrades

Substation	Facility Owner	Cost (\$)
SUNRISE	Alliant Energy	112,000
WILCOX	Alliant Energy	212,000

Table 9.2-6 – PJM Affected System Upgrades

Constraint	Facility Owner	Network Upgrade	Cost (\$)
GARDEN PR; R - SILVER LK; R Ckt 1 345 kV line	ComEd	Upgrade GARDEN PR; R - SILVER LK; R Ckt 1 345 kV line	32,400,000
CHERRY VA; B - GARDEN PR; R Ckt 1 345 kV line	ComEd	Install new 345kV line from the AB1-089 interconnection substation to TSS 144 Wayne station	165,000,000

Table 9.2-7 – ERIS Shared Network Upgrades

Network Upgrade	Facility Owner	ISD (For Cost Estimate Only)	Cost (\$)	Self-Fund
None	-	-	-	-

9.3 NRIS Network Upgrades Proposed for DPP 2019 Cycle 1 Phase 2 Projects

Network upgrades for Network Resource Interconnection Service (NRIS) were identified in the MISO's deliverability analysis and listed in the Table 9.3-1 below.

Table 9.3-1 – NRIS Network Upgrades Identified

Network Upgrade	Facility Owner	ISD (For Cost Estimate Only) ¹	Cost Used for NRIS Cost Allocation (\$) ²	Self-Fund
Neevin – Butte Des Morts 138 kV (43021), Rebuild	ATC	10/01/2021	1,556,274	No
University – Mukwonago 138 kV (UNIG52), Rebuild	ATC	12/01/2021	31,814,057	No
Jefferson – Crawfish River 138 kV (6654), Rebuild	ATC	7/15/2022	3,147,000	No
North Fond Du Lac – Aviation 138 kV (G111), Rebuild	ATC	10/01/2021	12,443,358	No
Aviation - Ellinwood 138 kV (X-50), Rebuild	ATC	10/01/2021	2,807,900	No
Nine Mile – Pine River 69 kV (6923/6921), Double Circuit Rebuild	ATC	3/01/2021	25,477,000	No
Rock River - Marine Tap 138 kV (X-23), Partial Reconductor	ATC	12/01/2021	1,098,413	No
Barneveld Military Ridge Interconnection Project (MTEP PID:18052)	ATC	01/01/2024	0	NA
Eden - Highland 138 kV (X-147), Reconductor	ATC	08/01/2023	3,570,774	No
Fox River – North Appleton 345 kV (NAPL121), Uprate	ATC	10/30/2022	292,629	No
North Lake Geneva – North Lake Geneva Tap 138 kV (6541), Uprate	ATC	12/01/2021	411,985	No
North Lake Geneva Tap – Burlington 138 kV (6541), Reconductor	ATC	12/01/2021	345,423	No
White Water – Bluff Creek – Sugar Creek 138 kV (WHIG53), Uprate	ATC	12/01/2021	399,588	No
Elkhorn – North Lake Geneva 138 kV (X-55), Uprate	ATC	12/01/2021	6,859	No
Mass – Bruce Crossing – Watersmeet 69 kV (6530), Uprate	ATC	9/15/2021	964,627	No
Columbia – South Fond du Lac 345 kV (W-5), Uprate	ATC	8/01/2021	578,748	No
Jefferson 138 kV, Replace Switches	ATC	7/15/2022	194,675	No
Crawfish River – Concord 138 kV (9043), Uprate	ATC	7/15/2022	1,181,456	No
J986 POI – Port Edwards (X-11) 138 kV, Uprate	ATC	10/01/2021	4,068,174	No

¹ All Network Upgrades were estimated on the earliest ISD dollars of responsible generators.

² ATC Network Upgrades are estimated in ISD dollars with a 2.5% annual escalation rate and include a contingency based on a project risk generally between 5% – 20%.

9.4 Cost Allocation Methodology for Thermal Network Upgrades

The costs of Network Upgrades (NU) for a set of generation projects (one or more subgroups or entire group with identified NU) are based off the MW impact of the worst-case scenario for each specific generator project. Basically, whatever the highest MW impact (increasing flow) is for that particular generator where the constraint is identified and requires NU is how it should be calculated.

Constraints which are mitigated by one or a subset of NU are identified. The highest MW contribution on these constraints from each generating facility is calculated in the MISO DPP study models without any Network Upgrades. Then the cost of each NU is allocated based on the pro rata share of the MW contribution from each generating facility on the constraints mitigated or partly mitigated by this NU. The methodology to determine the cost allocation of NU is:

$$\text{Project A cost portion of NU} = \text{Cost of NU} \times \left(\frac{\text{Max}(\text{Proj. A MW contribution on constraint})}{\sum_i \text{Max}(\text{Proj. i MW contribution on constraint})} \right)$$

9.5 Cost Allocation Methodology for Short Circuit Upgrades

For each ATC breaker shown to be overdutied, a short circuit network upgrade will be scoped to address the overduty issue and the cost of that upgrade will be assigned to generators. No breaker duty analysis is performed for Affected System equipment. Appendix N provides short circuit data for all substation buses 69 kV and above included in ATC protection CAPE. Any non-ATC equipment requiring a change would be considered Affected System.

Once costs of short circuit network upgrades are determined, they are allocated proportionally to study generators that have a greater than 3% of the total of all current queue study generator contributing fault currents under the single line-ground short circuit fault simulation at the identified substation.

The system impact study does not include any substation ground grid screening study. ATC conducts a ground grid evaluation for ATC brownfield substations in MISO DPP Facility Studies based on available ground grid design documents or in-house models when major substation work is identified. Major substation work includes, but is not limited to:

- Generation addition (brownfield POIs)
- Substation expansion
- Transmission line addition
- Transformer addition or replacement
- Protective device replacement for SLG short-circuit duty reasons

9.6 Cost Allocation Tables

The cost allocation of Network Upgrades for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of Energy Resource Interconnection Service or Network Resource Interconnection service as of the issued date of the System Impact Study report.

Assuming all generating facilities in the DPP 2019 Cycle 1 Phase 2 (East ATC) group advance, Table 9.6-1, Table 9.6-2, Table 9.6-3, Table 9.6-4, Table 9.6-5, and Table 9.6-6 show how the costs for ERIS steady-state Network Upgrades, Alliant Energy Affected System Network Upgrades, PJM Affected System Network Upgrades, and NRIS Network Upgrades are allocated to responsible generating facilities.

Table 9.6-1 – ERIS Steady-State Network Upgrade Responsible Generator Cost Allocation

Required ERIS Network Upgrades	Cost Estimates (\$)	Worst MW Impact		Total MW Impact	Cost Allocation %		Cost Allocation (\$)	
		J1244	J1370		J1244	J1370	J1244	J1370
Dafter – Nine Mile (ESE_6957) 69 kV, Bifurcate	325,098	-	50.00	50.00	-	100.00	-	325,098
Winona - Atlantic (Winona69) 69kV, Reconductor; MTEP PID:4727	0	38.40	-	38.40	100.00	-	0	-
Total ERIS Steady State Thermal Network Upgrade Cost (\$) Allocated to Each Generator							0	325,098

Table 9.6-2 – WPL Affected System Upgrades

Substation	Cost Estimates (\$)	Cost Allocation %	Cost Allocation (\$)
		J1460	J1460
SUNRISE	112,000	100	112,000
WILCOX	212,000	100	212,000
Total ERIS WPL Affected System Upgrade Cost (\$) Allocated to Each Generator			324,000

Table 9.6-3 – PJM Affected System Upgrades

Network Upgrade	Total Cost (\$)	PJM Affected System Costs Allocated (\$)	
		J1214	J1377
Upgrade GARDEN PR; R - SILVER LK; R Ckt 1 345 kV line	32,400,000	3,270,000	1,410,000
Install new 345kV line from the AB1-089 interconnection substation to TSS 144 Wayne station	165,000,000	86,150,000	37,310,000
Total Network Upgrade Costs (\$)		89,420,000	38,720,000

Table 9.6-4 – NRIS Network Upgrade Responsible Generator Worst MW Impact Data

Required NRIS Network Upgrades	Cost Used for NRIS Network Upgrade Allocation (\$)	Worst MW Impact															Total MW Impact
		J1214	J1244	J1251	J1253	J1304	J1305	J1316	J1326	J1345	J1370	J1374	J1377	J1410	J1411	J1460	
Neevin – Butte Des Morts 138 kV (43021), Rebuild	1,556,274	-	-	-	16.25	-	-	-	-	2.05	-	-	-	15.52	3.88	-	37.7
University – Mukwonago 138 kV (UNIG52), Rebuild	31,814,057	-	-	-	-	5.18	3.76	-	-	-	-	-	-	-	-	33.04	41.98
Jefferson – Crawfish River 138 kV (6654), Rebuild	3,147,000	18.62	-	-	-	-	-	-	4.65	-	-	-	5.2	15.34	3.83	-	47.64
North Fond Du Lac – Aviation 138 kV (G111) , Rebuild	12,443,358	-	-	-	-	-	-	-	-	2.34	-	-	-	-	-	-	2.34
Aviation - Ellinwood 138 kV (X-50), Rebuild	2,807,900	-	-	-	-	-	-	-	-	2.34	-	-	-	-	-	-	2.34
Nine Mile – Pine River 69 kV (6923/6921), Double Circuit Rebuild	25,477,000	-	-	-	-	-	-	-	-	-	35.98	-	-	-	-	-	35.98
Rock River - Marine Tap 138 kV (X-23), Partial Reconductor	1,098,413	-	-	-	-	4.69	6.63	-	-	-	-	-	-	-	-	29.32	40.64
Barneveld Military Ridge Interconnection Project (MTEP PID:18052)	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Eden - Highland 138 kV (X-147), Reconductor	3,570,774	-	-	-	-	-	-	-	-	-	-	10.51	-	-	-	-	10.51
Fox River – North Appleton 345 kV (NAPL121), Uprate	292,629	-	-	-	-	-	-	2.97	-	-	-	-	-	-	-	-	2.97
North Lake Geneva – North Lake Geneva Tap 138 kV (6541), Uprate	411,985	-	-	-	-	5.99	4.16	-	-	-	-	-	-	-	-	19.6	29.75
North Lake Geneva Tap – Burlington 138 kV (6541), Reconductor	345,423	-	-	-	-	6.87	4.9	-	-	-	-	-	-	-	-	27.82	39.59
White Water – Bluff Creek – Sugar Creek 138 kV (WHIG53), Uprate	399,588	-	-	-	-	3.72	2.79	-	-	-	-	-	-	-	-	32.24	38.75
Elkhorn – North Lake Geneva 138 kV (X-55), Uprate	6,859	-	-	-	-	8.4	5.92	-	-	-	-	-	-	-	-	28.18	42.5
Mass – Bruce Crossing – Watersmeet 69 kV (6530), Uprate	964,627	-	4.98	-	-	-	-	-	-	-	-	-	-	-	-	-	4.98
Columbia – South Fond du Lac 345 kV (W-5), Uprate	578,748	34.92	-	-	-	3.81	3.24	-	8.73	-	-	6.54	9.8	57.3	14.33	-	138.67
Jefferson 138 kV, Replace Switches	194,675	18.84	-	-	-	-	-	-	4.71	-	-	-	5.31	-	-	-	28.86
Crawfish River – Concord 138 kV (9043), Uprate	1,181,456	18.62	-	-	-	-	-	-	4.65	-	-	-	5.2	15.34	3.83	-	47.64
J986 POI – Port Edwards (X-11) 138 kV, Uprate	4,068,174	-	-	-	-	-	-	-	-	1.28	-	-	-	-	-	-	1.28

Table 9.6-5 – NRIS Network Upgrade Responsible Generator Cost Allocation Percentage Data

Required NRIS Network Upgrades	Cost Used for NRIS Network Upgrade Allocation (\$)	Cost Allocation %														
		J1214	J1244	J1251	J1253	J1304	J1305	J1316	J1326	J1345	J1370	J1374	J1377	J1410	J1411	J1460
Neevin – Butte Des Morts 138 kV (43021), Rebuild	1,556,274	-	-	-	43.10	-	-	-	-	5.44	-	-	-	41.17	10.29	-
University – Mukwonago 138 kV (UNIG52), Rebuild	31,814,057	-	-	-	-	12.34	8.96	-	-	-	-	-	-	-	-	78.70
Jefferson – Crawfish River 138 kV (6654), Rebuild	3,147,000	39.08	-	-	-	-	-	-	9.76	-	-	-	10.92	32.20	8.04	-
North Fond Du Lac – Aviation 138 kV (G111), Rebuild	12,443,358	-	-	-	-	-	-	-	-	100.00	-	-	-	-	-	-
Aviation - Ellinwood 138 kV (X-50), Rebuild	2,807,900	-	-	-	-	-	-	-	-	100.00	-	-	-	-	-	-
Nine Mile – Pine River 69 kV (6923/6921), Double Circuit Rebuild	25,477,000	-	-	-	-	-	-	-	-	-	100.00	-	-	-	-	-
Rock River - Marine Tap 138 kV (X-23), Partial Reconductor	1,098,413	-	-	-	-	11.54	16.31	-	-	-	-	-	-	-	-	72.15
Barneveld Military Ridge Interconnection Project (MTEP PID:18052)	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Eden - Highland 138 kV (X-147), Reconductor	3,570,774	-	-	-	-	-	-	-	-	-	-	100.00	-	-	-	-
Fox River – North Appleton 345 kV (NAPL121), Uprate	292,629	-	-	-	-	-	-	100.00	-	-	-	-	-	-	-	-
North Lake Geneva – North Lake Geneva Tap 138 kV (6541), Uprate	411,985	-	-	-	-	20.13	13.99	-	-	-	-	-	-	-	-	65.88
North Lake Geneva Tap – Burlington 138 kV (6541), Reconductor	345,423	-	-	-	-	17.35	12.38	-	-	-	-	-	-	-	-	70.27
White Water – Bluff Creek – Sugar Creek 138 kV (WHIG53), Uprate	399,588	-	-	-	-	9.60	7.20	-	-	-	-	-	-	-	-	83.20
Elkhorn – North Lake Geneva 138 kV (X-55), Uprate	6,859	-	-	-	-	19.76	13.93	-	-	-	-	-	-	-	-	66.31
Mass – Bruce Crossing – Watersmeet 69 kV (6530), Uprate	964,627	-	100.00	-	-	-	-	-	-	-	-	-	-	-	-	-
Columbia – South Fond du Lac 345 kV (W-5), Uprate	578,748	25.18	-	-	-	2.75	2.34	-	6.30	-	-	4.72	7.07	41.31	10.33	-
Jefferson 138 kV, Replace Switches	194,675	65.28	-	-	-	-	-	-	16.32	-	-	-	18.40	-	-	-
Crawfish River – Concord 138 kV (9043), Uprate	1,181,456	39.08	-	-	-	-	-	-	9.76	-	-	-	10.92	32.20	8.04	-
J986 POI – Port Edwards (X-11) 138 kV, Uprate	4,068,174	-	-	-	-	-	-	-	-	100.00	-	-	-	-	-	-

Table 9.6-6 – NRIS Network Upgrade Responsible Generator Cost Allocations

Required NRIS Network Upgrades	Cost Used for NRIS Network Upgrade Allocation (\$)	Cost Allocation (\$)														
		J1214	J1244	J1251	J1253	J1304	J1305	J1316	J1326	J1345	J1370	J1374	J1377	J1410	J1411	J1460
Neevin – Butte Des Morts 138 kV (43021), Rebuild	1,556,274	-	-	-	670,754	-	-	-	-	84,661	-	-	-	640,718	160,141	-
University – Mukwonago 138 kV (UNIG52), Rebuild	31,814,057	-	-	-	-	3,925,855	2,850,540	-	-	-	-	-	-	-	-	25,037,663
Jefferson – Crawfish River 138 kV (6654), Rebuild	3,147,000	1,229,848	-	-	-	-	-	-	307,147	-	-	-	343,652	1,013,334	253,019	-
North Fond Du Lac – Aviation 138 kV (G111), Rebuild	12,443,358	-	-	-	-	-	-	-	-	12,443,358	-	-	-	-	-	-
Aviation - Ellinwood 138 kV (X-50), Rebuild	2,807,900	-	-	-	-	-	-	-	-	2,807,900	-	-	-	-	-	-
Nine Mile – Pine River 69 kV (6923/6921), Double Circuit Rebuild	25,477,000	-	-	-	-	-	-	-	-	-	25,477,000	-	-	-	-	-
Rock River - Marine Tap 138 kV (X-23), Partial Reconductor	1,098,413	-	-	-	-	126,757	179,151	-	-	-	-	-	-	-	-	792,505
Barneveld Military Ridge Interconnection Project (MTEP PID:18052)	0	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Eden - Highland 138 kV (X-147), Reconductor	3,570,774	-	-	-	-	-	-	-	-	-	-	3,570,774	-	-	-	-
Fox River – North Appleton 345 kV (NAPL121), Uprate	292,629	-	-	-	-	-	-	292,629	-	-	-	-	-	-	-	-
North Lake Geneva – North Lake Geneva Tap 138 kV (6541), Uprate	411,985	-	-	-	-	82,933	57,637	-	-	-	-	-	-	-	-	271,416
North Lake Geneva Tap – Burlington 138 kV (6541), Reconductor	345,423	-	-	-	-	59,931	42,763	-	-	-	-	-	-	-	-	242,729
White Water – Bluff Creek – Sugar Creek 138 kV (WHIG53), Uprate	399,588	-	-	-	-	38,360	28,770	-	-	-	-	-	-	-	-	332,457
Elkhorn – North Lake Geneva 138 kV (X-55), Uprate	6,859	-	-	-	-	1,355	955	-	-	-	-	-	-	-	-	4,548
Mass – Bruce Crossing – Watersmeet 69 kV (6530), Uprate	964,627	-	964,627	-	-	-	-	-	-	-	-	-	-	-	-	-
Columbia – South Fond du Lac 345 kV (W-5), Uprate	578,748	145,729	-	-	-	15,916	13,543	-	36,461	-	-	27,317	40,917	239,081	59,785	-
Jefferson 138 kV, Replace Switches	194,675	127,084	-	-	-	-	-	-	31,771	-	-	-	35,820	-	-	-
Crawfish River – Concord 138 kV (9043), Uprate	1,181,456	461,713	-	-	-	-	-	-	115,310	-	-	-	129,015	380,429	94,989	-
J986 POI – Port Edwards (X-11) 138 kV, Uprate	4,068,174	-	-	-	-	-	-	-	-	4,068,174	-	-	-	-	-	-
Total NRIS Network Upgrade Cost (\$) Allocated to Each Generator		1,964,373	964,627	0	670,754	4,251,106	\$3,173,359	292,629	490,689	19,404,093	25,477,000	3,598,091	549,405	2,273,562	567,933	26,681,318

10.0 AVAILABLE APPENDIX DOCUMENTS (NOT ATTACHED)

Appendix A – Study Criteria, Methodology, and Assumptions

Appendix B – ATC Planning Criteria and Generation Facility Interconnection Guide

Appendix C – Interconnection Facility Project Diagrams_CEI

Appendix D – Network Upgrade Project Diagrams_CEI

Note: Project Diagrams were not developed for line upgrade projects.

Appendix E – Steady State Power Flow Results_CEI

Appendix F – Steady State Operating Restriction Study Results_CEI

Appendix G – Dynamic Stability Results_CEI

Appendix H – PJM Affected System Study Results_CEI

Appendix I – MISO Deliverability Study Results_CEI

Appendix J – Assessed System Performance Reference

Appendix L – PSCAD Study Results_CEI

Appendix M – Conditions to GIA

Appendix N - Short Circuit Data_CEI